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Image forming apparatus having a toner replenishment operation

Abstract

An image forming apparatus to and from which a replenishment container accommodating toner is attachable and detachable includes a replenishment port to allow replenishment of toner from the replenishment container, an opening/closing portion to open the replenishment port in an open position and close the replenishment port in a closed position, a locking member to move between a restricting position in which the locking member restricts movement of the opening/closing portion from the closed position to the open position, and an allowing position in which the movement of the opening/closing portion from the closed position to the open position is allowed. A developer container accepts more toner for replenishment in a case where a second display is in a second state than in a first state, and a first display more accurately displays information related to the toner accommodated in the developer container than the second display.

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Background/Summary

(1) This application is a continuation of application Ser. No. 17/586,857, filed Jan. 28, 2022, which is a continuation of application Ser. No. 17/034,387, filed Sep. 28, 2020, now U.S. Pat. No. 11,262,683, issued Mar. 1, 2022.

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to an image forming apparatus that forms an image on a recording material.
Description of the Related Art
(2) Typically, an image forming apparatus of an electrophotographic system forms an image on a recording material by developing an electrostatic latent image formed on the surface of a photosensitive member into a toner image by using toner, and then transferring the toner image from the photosensitive member onto the recording material. As methods for replenishing an image forming apparatus with toner consumed by repetitively performing image formation, a process cartridge system and a consecutive replenishment system are known. The process cartridge system is a system in which a photosensitive member and a developer container accommodating toner are integrated as a process cartridge, and the process cartridge is replaced by a brand-new one when all toner in the developer container is consumed.
(3) Meanwhile, Japanese Patent Laid-Open No. H08-30084 discloses a developing unit of a consecutive replenishment system that includes a toner conveyance path through which toner is supplied to a developing roller, and a developer supply box connected to the toner conveyance path, and that supplies toner from the developer supply box to the toner conveyance path in accordance with a detection result of a toner remainder amount.
(4) In recent years, demand from users for a wider variety of use of the image forming apparatus

has been increasing in addition to the process cartridge system and the consecutive replenishment system described above.

SUMMARY OF THE INVENTION

(5) According to a first aspect of the present invention, an image forming apparatus to and from which a replenishment container accommodating toner is attachable and detachable and which is configured to form an image on a recording material, the image forming apparatus includes an image bearing member, a developer container configured to accommodate toner, a developing portion configured to develop an electrostatic image formed on the image bearing member into a toner image by using the toner accommodated in the developer container, a replenishment port configured to allow replenishment of toner from the replenishment container, which is arranged outside of the image forming apparatus, to the developer container therethrough in a state where the replenishment container is attached to the replenishment port, a first display portion configured to display a ratio of an amount of toner accommodated in the developer container to a maximum amount of toner that the developer container is capable of accommodating, a second display portion configured to switch between a first state and a second state different from the first state, the developer container being capable of accepting more toner for replenishment in a case where the second display portion is in the second state than in a case where the second display portion is in the first state, and a controller configured to, in a case where a replenishment operation, in which toner is supplied from the replenishment container to the replenishment port, is performed when the second display portion is in the second state, switch the second display portion from the second state to the first state and perform a display processing of displaying, on the first display portion, the ratio corresponding to an amount of toner accommodated in the developer container after the replenishment operation.

(6) According to a second aspect of the present invention, an image forming apparatus to and from which a replenishment container accommodating toner is attachable and detachable and which is communicable with an information processing apparatus including a first display portion and is configured to form a toner image on a recording material, the image forming apparatus includes an image bearing member, a developer container configured to accommodate toner, a developing portion configured to develop an electrostatic image formed on the image bearing member into a toner image by using the toner accommodated in the developer container, a replenishment port configured to allow replenishment of toner from the replenishment container, which is arranged outside of the image forming apparatus, to the developer container therethrough in a state where the replenishment container is attached to the replenishment port, a second display portion configured to switch between a first state and a second state different from the first state, the developer container being capable of accepting more toner for replenishment in a case where the second display portion is in the second state than in a case where the second display portion is in the first state, a controller configured to, in a case where a replenishment operation, in which toner is supplied from the replenishment container to the replenishment port, is performed when the second display portion is in the second state, switch the second display portion from the second state to the first state and perform a display processing of displaying, on the first display portion, a ratio of an amount of toner accommodated in the developer container after the replenishment operation to a maximum amount of toner that the developer container is capable of accommodating.

(7) According to a third aspect of the present invention, an image forming apparatus to and from which a replenishment container accommodating toner is attachable and detachable and which is configured to form an image on a recording material, the image forming apparatus includes an image bearing member, a developer container configured to accommodate toner, a developing portion configured to develop an electrostatic image formed on the image bearing member into a toner image by using the toner accommodated in the developer container, a replenishment port configured to allow replenishment of toner from the replenishment container, which is arranged outside of the image forming apparatus, to the developer container therethrough in a state where the

replenishment container is attached to the replenishment port, a first display portion configured to display a printable sheet number, a second display portion configured to switch between a first state and a second state different from the first state, the developer container being capable of accepting more toner for replenishment in a case where the second display portion is in the second state than in a case where the second display portion is in the first state, and a controller configured to, in a case where a replenishment operation, in which toner is supplied from the replenishment container to the replenishment port, is performed when the second display portion is in the second state, switch the second display portion from the second state to the first state and perform a display processing of displaying, on the first display portion, the printable sheet number corresponding to an amount of toner accommodated in the developer container after the replenishment operation.
(8) According to a fourth aspect of the present invention, an image forming apparatus to and from which a replenishment container accommodating toner is attachable and detachable and which is communicable with an information processing apparatus including a first display portion and is configured to form a toner image on a recording material, the image forming apparatus includes an image bearing member, a developer container configured to accommodate toner, a developing portion configured to develop an electrostatic image formed on the image bearing member into a toner image by using the toner accommodated in the developer container, a replenishment port configured to allow replenishment of toner from the replenishment container, which is arranged outside of the image forming apparatus, to the developer container therethrough in a state where the replenishment container is attached to the replenishment port, a second display portion configured to switch between a first state and a second state different from the first state, the developer container being capable of accepting more toner for replenishment in a case where the second display portion is in the second state than in a case where the second display portion is in the first state, and a controller configured to, in a case where a replenishment operation, in which toner is supplied from the replenishment container to the replenishment port, is performed when the second display portion is in the second state, switch the second display portion from the second state to the first state and perform a display processing of displaying, on the first display portion, a printable sheet number corresponding to the amount of toner accommodated in the developer container after the replenishment operation.
(9) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a section view of an image forming apparatus according to a first embodiment.
(2) FIG. **1**B is a perspective view of the image forming apparatus according to the first embodiment.
(3) FIG. **2**A is a section view of the image forming apparatus according to the first embodiment.
(4) FIG. **2**B is a perspective view of the image forming apparatus according to the first embodiment.
(5) FIG. **3** is a diagram for describing attachment and detachment of a process cartridge according to the first embodiment.
(6) FIG. **4**A is a diagram for describing an openable and closable member of the image forming apparatus according to the first embodiment.
(7) FIG. **4**B is a diagram for describing the openable and closable member of the image forming apparatus according to the first embodiment.
(8) FIG. **4**C is a diagram for describing the openable and closable member of the image forming apparatus according to the first embodiment.

(9) FIG. **5**A is a diagram for describing a configuration of the process cartridge according to the first embodiment.
(10) FIG. **5**B is a diagram for describing the configuration of the process cartridge according to the first embodiment.
(11) FIG. **6**A is a diagram for describing the configuration of the process cartridge according to the first embodiment.
(12) FIG. **6**B is a diagram for describing the configuration of the process cartridge according to the first embodiment.
(13) FIG. **6**C is a diagram for describing the configuration of the process cartridge according to the first embodiment.
(14) FIG. **7**A is a perspective view of a toner pack according to the first embodiment.
(15) FIG. **7**B is a side view of the toner pack according to the first embodiment.
(16) FIG. **8**A is a perspective view of the toner pack according to the first embodiment.
(17) FIG. **8**B is a side view of the toner pack according to the first embodiment.
(18) FIG. **8**C is a diagram illustrating how toner is discharged.
(19) FIG. **9**A is a perspective view of a replenishment container attaching portion according to the first embodiment.
(20) FIG. **9**B is a top view of the replenishment container attaching portion according to the first embodiment.
(21) FIG. **9**C is an enlarged view of the replenishment container attaching portion according to the first embodiment.
(22) FIG. **10**A is a diagram for describing an operation of the replenishment container attaching portion according to the first embodiment.
(23) FIG. **10**B is a diagram for describing the operation of the replenishment container attaching portion according to the first embodiment.
(24) FIG. **11**A is a diagram illustrating a position of a locking member according to the first embodiment.
(25) FIG. **11**B is a diagram illustrating a position of the locking member according to the first embodiment.
(26) FIG. **12** is a perspective view of the toner pack according to the first embodiment.
(27) FIG. **13** is a diagram illustrating a pressing mechanism of the locking member according to the first embodiment.
(28) FIG. **14**A is a diagram illustrating a panel according to the first embodiment.
(29) FIG. **14**B is a diagram illustrating the panel according to the first embodiment.
(30) FIG. **14**C is a diagram illustrating the panel according to the first embodiment.
(31) FIG. **15**A is a perspective view of a toner bottle unit according to a first modification example.
(32) FIG. **15**B is a perspective view of the toner bottle unit according to the first modification example.
(33) FIG. **15**C is a side view of the toner bottle unit according to the first modification example.
(34) FIG. **15**D is a section view of the toner bottle unit according to the first modification example.
(35) FIG. **16**A is a diagram for describing an inner configuration of the toner bottle unit according to the first modification example.
(36) FIG. **16**B is a diagram for describing the inner configuration of the toner bottle unit according to the first modification example.
(37) FIG. **16**C is a diagram for describing the inner configuration of the toner bottle unit according to the first modification example.
(38) FIG. **16**D is a diagram for describing the inner configuration of the toner bottle unit according to the first modification example.
(39) FIG. **16**E is a diagram for describing detection of rotation of the toner bottle unit.
(40) FIG. **16**F is a diagram for describing detection of rotation of the toner bottle unit.

(41) FIG. **17**A is a perspective view of a process cartridge according to a second modification example.
(42) FIG. **17**B is a top view of the process cartridge according to the second modification example.
(43) FIG. **17**C is a section view of the process cartridge according to the second modification example.
(44) FIG. **17**D is a section view of the process cartridge according to the second modification example.
(45) FIG. **18**A is a perspective view of a process cartridge according to a third modification example.
(46) FIG. **18**B is a top view of the process cartridge according to the third modification example.
(47) FIG. **18**C is a section view of the process cartridge according to the third modification example.
(48) FIG. **19** is a block diagram illustrating a control system of the image forming apparatus according to the first embodiment.
(49) FIG. **20** is a perspective view of a personal computer and a mobile information processing terminal connected to an image forming apparatus.
(50) FIG. **21**A is a perspective view of a panel in a first state.
(51) FIG. **21**B is a perspective view of the panel in a second state.
(52) FIG. **21**C is a perspective view of the panel in a third state.
(53) FIG. **21**D is a perspective view of the panel in a fourth state.
(54) FIG. **22** is a flowchart illustrating control performed when replenishing toner.
(55) FIG. **23** is a graph showing the amount of toner in a developer container.
(56) FIG. **24**A is a graph showing the amount of toner in the developer container.
(57) FIG. **24**B is a perspective view of a display portion displaying a message.
(58) FIG. **25**A is a graph showing the amount of toner in a developer container according to a second embodiment.
(59) FIG. **25**B is a perspective view of a display portion displaying a message.
(60) FIG. **26**A is a graph showing the amount of toner in a developer container according to a third embodiment.
(61) FIG. **26**B is a perspective view of a display portion displaying a message.
(62) FIG. **27** is a perspective view of a panel according to a fourth embodiment.
(63) FIG. **28**A is a graph showing the amount of toner in a developer container according to a fourth embodiment.
(64) FIG. **28**B is a perspective view of a display portion displaying a message.

DESCRIPTION OF THE EMBODIMENTS

(65) Exemplary embodiments of the present invention will be described below with reference to drawings.

First Embodiment

(1) Image Forming Apparatus

(66) FIG. **1**A is a schematic diagram illustrating a configuration of an image forming apparatus **1** according to a first embodiment. The image forming apparatus **1** is a monochromatic printer that forms an image on a recording material on the basis of image information input from an external device. Examples of the recording material include sheet materials of different natures. Examples of the sheet materials include paper sheets such as regular paper sheets and cardboards, plastic films such as sheets for overhead projectors, sheets having irregular shapes such as envelops and index sheets, and cloths.

(67) (1-1) Overall Configuration

(68) As illustrated in FIGS. **1**A and **1**B, the image forming apparatus **1** includes a printer body **100** serving as an apparatus body, a reading apparatus **200** openably and closably supported on the printer body **100**, and an operation portion **300** attached to an exterior surface of the printer body

100. The printer body **100** includes an image forming portion **10**, a feeding portion **60**, a fixing portion **70**, and a discharge roller pair **80**. The feeding portion **60** feeds a recording material to the image forming portion **10**, and the image forming portion **10** forms a toner image on the recording material. The fixing portion **70** fixes the toner image formed by the image forming portion **10** onto the recording material, and the discharge roller pair **80** discharges the recording material having passed through the fixing portion **70** to the outside of the apparatus. In addition, a direct replenishment system in which toner is directly replenished from the outside of the image forming apparatus **1** by using a toner pack **40** filled with toner for replenishment is employed for a process cartridge **20** of the present embodiment.
(69) The image forming portion **10** is an image forming portion of an electrophotographic system including a scanner unit **11**, the process cartridge **20**, and a transfer roller **12**. The process cartridge **20** includes a photosensitive drum **21**, a charging roller **22** disposed in the vicinity of the photosensitive drum **21**, a developing roller **31**, and a cleaning blade **24**.
(70) The photosensitive drum **21** serving as an image bearing member of the present embodiment is a photosensitive member formed in a cylindrical shape. The photosensitive drum **21** of the present embodiment includes a drum-shaped base body formed from aluminum, and a photosensitive layer formed from a negatively-chargeable organic photoconductor on the base body. In addition, the photosensitive drum **21** is rotationally driven by a motor at a predetermined process speed in a predetermined direction, which is a clockwise direction in FIG. **1**A.
(71) The charging roller **22** comes into contact with the photosensitive drum **21** at a predetermined pressure contact force, and thus forms a charging portion. In addition, a desired charging voltage is applied to the charging roller **22** from a charging high-voltage power source, and thus the charging roller **22** uniformly charges the surface of the photosensitive drum **21** to a predetermined potential. In the present embodiment, the photosensitive drum **21** is negatively charged by the charging roller **22**.
(72) The scanner unit **11** radiates laser light L corresponding to image information input from an external device or the reading apparatus **200** onto the photosensitive drum **21** by using a polygonal mirror, and thus exposes the surface of the photosensitive drum **21** in a scanning manner. As a result of this exposure, an electrostatic latent image corresponding to the image information is formed on the surface of the photosensitive drum **21**. To be noted, the scanner unit **11** is not limited to a laser scanner unit. For example, a light-emitting diode: LED exposing unit including an LED array in which a plurality of LEDs are arranged along the longitudinal direction of the photosensitive drum **21** may be employed.
(73) A developing unit **802** includes a developing roller **31** serving as a developer bearing member configured to bear a developer, a developer container **32** serving as a frame member of the developing unit **802**, and a supply roller **33** capable of supplying the developer to the developing roller **31**. The developing roller **31** and the supply roller **33** are rotatably supported by the developer container **32**. In addition, the developing roller **31** is disposed in an opening portion of the developer container **32** so as to oppose the photosensitive drum **21**. The supply roller **33** is rotatably in contact with the developing roller **31**, and toner serving as the developer accommodated in the developer container **32** is applied on the surface of the developing roller **31** by the supply roller **33**. The developer container is also called a developer storage container.
(74) The developing unit **802** of the present embodiment employs a contact developing system as a developing system. That is, a toner layer born on the developing roller **31** serving as a developing portion comes into contact with the photosensitive drum **21** in a developing portion serving as a developing region where the photosensitive drum **21** and the developing roller **31** oppose each other. A developing voltage is applied to the developing roller **31** from a developing high-voltage power source. Under the influence of the developing voltage, the toner born on the developing roller **31** transfers from the developing roller **31** onto the surface of the photosensitive drum **21** in accordance with the potential distribution of the surface of the photosensitive drum **21**, and thus the

electrostatic latent image is developed into a toner image. To be noted, in the present embodiment, a reversal development system is employed. That is, the toner image is formed by the toner attaching to a region where the amount of charge is reduced by being exposed in an exposing step on the surface of the photosensitive drum **21** charged in a charging step.
(75) In addition, in the present embodiment, toner which has a particle diameter of 6 μm and whose normal charging polarity is a negative polarity is used. For example, a polymer toner generated by a polymerization method is employed as the toner of the present embodiment. In addition, the toner of the present embodiment is a so-called nonmagnetic one-component developer that does not contain a magnetic component, and is born on the developing roller **31** mainly by an intermolecular force and an electrostatic force, that is, an image force. However, a one-component developer containing a magnetic component may be used. In addition, in some cases, the one-component developer contains additives for adjusting the fluidity and charging performance of the toner in addition to the toner particles. Examples of the additives include wax and silica fine particles. In addition, a two-component developer constituted by a nonmagnetic toner and a magnetic carrier may be used as the developer. In the case of using a magnetic developer, a cylindrical developing sleeve in which a magnet is disposed is used as the developer bearing member. That is, the developer contained in the developer container **32** is not limited to a one-component developer containing only a toner component, and may be a two-component developer containing toner and carrier.
(76) An agitation member **34** serving as an agitation portion is provided inside the developer container **32**. The agitation member **34** is driven to pivot, and thus agitates the toner in the developer container **32** and conveys the toner toward the developing roller **31** and the supply roller **33**. In addition, the agitation member **34** has a function of circulating toner not used for development and peeled off from the developing roller **31** in the developer container **32**, and thus making the toner in the developer container **32** uniform.
(77) In addition, a developing blade **35** that regulates the amount of toner born on the developing roller **31** is disposed at an opening portion of the developer container **32** where the developing roller **31** is disposed. In accordance with the rotation of the developing roller **31**, the toner supplied to the surface of the developing roller **31** passes through a portion where the developing roller **31** and the developing blade **35** oppose each other, thus forms a uniform thin layer, and is negatively charged as a result of frictional charging.
(78) The feeding portion **60** includes a front door **61** supported to be openable and closable with respect to the printer body **100**, a supporting tray **62**, an inner plate **63**, a tray spring **64**, and a pickup roller **65**. The supporting tray **62** constitutes a bottom surface of a recording material accommodating space exposed by opening the front door **61**, and the inner plate **63** is supported on the supporting tray **62** so as to be capable of ascending and descending. The tray spring **64** urges the inner plate **63** upward, and presses a recording material P supported on the inner plate **63** against the pickup roller **65**. To be noted, the front door **61** closes the recording material accommodating space in the state of being closed with respect to the printer body **100**, and supports the recording material P together with the supporting tray **62** and the inner plate **63** in the state of being open with respect to the printer body **100**.
(79) The transfer roller **12** serving as a transfer portion transfers the toner image formed on the photosensitive drum **21** of the process cartridge **20** onto the recording material. To be noted, although a direct transfer system in which the toner image formed on the image bearing member is directly transferred from the image bearing member onto the recording material will be described in the present embodiment, an intermediate transfer system in which the toner image is transferred from the image bearing member via an intermediate transfer member such as an intermediate transfer belt may be employed. In that case, for example, a transfer unit constituted by an intermediate transfer belt, a primary transfer roller that transfers the toner image from the photosensitive drum onto the intermediate transfer belt through primary transfer, and a secondary

transfer roller that transfers the toner image from the intermediate transfer belt onto the recording material functions as a transfer portion.

(80) The fixing portion **70** is a thermal fixation system that performs an image fixing process by heating and melting the toner on the recording material. The fixing portion **70** includes a fixing film **71**, a fixing heater such as a ceramic heater that heats the fixing film **71**, a thermistor that measures the temperature of the fixing heater, and a pressurizing roller **72** that comes into pressure contact with the fixing film **71**.

(81) Next, an image forming operation of the image forming apparatus **1** will be described. When a command for image formation is input to the image forming apparatus **1**, an image forming process by the image forming portion **10** is started on the basis of image information input from an external computer connected to the image forming apparatus **1** or image information input from the reading apparatus **200**. The scanner unit **11** radiates laser light L toward the photosensitive drum **21** on the basis of the input image information. At this time, the photosensitive drum **21** has been charged by the charging roller **22** in advance, and an electrostatic latent image is formed on the photosensitive drum **21** by being irradiated with the laser light L. Then, this electrostatic latent image is developed by the developing roller **31**, and a toner image is formed on the photosensitive drum **21**.

(82) In parallel with the image forming process described above, the pickup roller **65** of the feeding portion **60** delivers out the recording material P supported on the front door **61**, the supporting tray **62**, and the inner plate **63**. The recording material P is fed to the registration roller pair **15** by the pickup roller **65**, and the skew thereof is corrected by abutting a nip of the registration roller pair **15**. In addition, the registration roller pair **15** is driven in accordance with a transfer timing of the toner image obtained from the start time of exposure performed by the scanner unit **11**, and conveys the recording material P to a transfer portion that is a nip portion formed between the transfer roller **12** and the photosensitive drum **21**.

(83) A transfer voltage is applied to the transfer roller **12** from the transfer high-voltage power source, and the toner image born on the photosensitive drum **21** is transferred onto the recording material P conveyed by the registration roller pair **15**. After the transfer, transfer residual toner on the surface of the photosensitive drum **21** is removed by the cleaning blade **24**, which is an elastic blade in contact with the photosensitive drum **21**. The recording material P onto which the toner image has been transferred is conveyed to the fixing portion **70** and passes through a nip portion formed between the fixing film **71** and the pressurizing roller **72** of the fixing portion **70**, and thus the toner image is heated and pressurized. As a result of this, the toner particles melt and then adhere to the recording material P. Thus, the toner image is fixed to the recording material P. The recording material P having passed through the fixing portion **70** is discharged to the outside of the image forming apparatus **1** by a discharge roller pair **80**, and is supported on a discharge tray **81** formed on an upper portion of the printer body **100**.

(84) The discharge tray **81** is inclined upward toward the downstream side in a discharge direction of the recording material, and trailing ends of recording materials discharged onto the discharge tray **81** are aligned by a regulating surface **84** by sliding down the discharge tray **81**.

(85) (1-2) Openable and Closable Part of Image Forming Apparatus

(86) As illustrated in FIGS. **2**A, **2**B, and **3**, a first opening portion **101** opening upward is provided in an upper portion of the printer body **100**. The first opening portion **101** is covered by a top cover **82** during use as illustrated in FIG. **1**B, and the process cartridge **20** is exposed by opening the top cover **82** upward as illustrated in FIG. **2**B. The top cover **82** is supported so as to be openable and closable with respect to the printer body **100** by rotating around a rotation shaft **82***c* illustrated in FIG. **3** extending in the left-right direction, and the discharge tray **81** is provided on the upper surface thereof. The top cover **82** is opened from the front side toward the rear side when the reading apparatus **200** is opened with respect to the printer body **100**. To be noted, the reading apparatus **200** and the top cover **82** are configured to be held in a state of being open and a state of being closed, by a holding mechanism such as a hinge mechanism.

(87) For example, the user opens the top cover **82** together with the reading apparatus **200** in the case where jam of the recording material has occurred in a conveyance path CP which the recording material fed by the pickup roller **65** passes through. Then, the user accesses the process cartridge **20** through the first opening portion **101** exposed by opening the top cover **82**, and pulls out the process cartridge **20** along a cartridge guide **102**. A projection portion **21***a* provided on an end portion of the process cartridge **20** in the axial direction of the photosensitive drum **21** illustrated in FIG. **5**A slides on the cartridge guide **102**, and thus the process cartridge **20** is guided by the cartridge guide **102**.
(88) Then, as a result of the process cartridge **20** being pulled out to the outside through the first opening portion **101**, a space through which a hand can reach the inside of the conveyance path CP is generated. The user can put their hand in the printer body **100** through the first opening portion **101** to access the recording material causing the jam in the conveyance path CP, and thus remove the recording material causing the jam.
(89) In addition, in the present embodiment, an opening/closing member **83** is openably and closably provided on the top cover **82** as illustrated in FIGS. **1**B and **4**C. An opening portion **82***a* opening upward is provided in the upper surface of the top cover **82** on which the discharge tray **81** is provided, and the opening portion **82***a* is covered by closing the opening/closing member **83**. The opening/closing member **83** and the opening portion **82***a* are provided on the right side of the op cover **82**. In addition, the opening/closing member **83** is supported on the top cover **82** so as to be openable and closable about a pivot shaft **83***a* extending in the front-rear direction, and is opened to the right by hooking a finger through a groove portion **82***b* provided on the top cover **82**. The opening/closing member **83** is formed in an approximately L-shape in accordance with the shape of the top cover **82**. To be noted, the opening/closing member **83** is not limited to the opening/closing mechanism described above. For example, the opening/closing member **83** may be disposed on the top cover **82** so as to cover a replenishment container attaching portion **701** and configured to open and close the opening portion **82***a* by sliding and pivoting on the upper surface of the top cover **82** about a pivot shaft perpendicular to the top cover **82**. Here, sliding on the upper surface of the top cover **82** means that the movement of the opening/closing member **83** in the pivot axis direction is restricted.
(90) The opening portion **82***a* is opened so as to expose the replenishment container attaching portion **701** provided in an upper portion of the process cartridge **20** for toner replenishment. By opening the opening/closing member **83**, the user can access the replenishment container attaching portion **701** without opening the top cover **82**. The user can replenish the process cartridge **20** with toner by attaching a toner pack **40** to the replenishment container attaching portion **701**.
(91) In the present embodiment, a system in which the user replenishes the process cartridge **20** with toner from the toner pack **40** filled with toner for replenishment illustrated in FIGS. **1**A and **1**B in a state in which the process cartridge **20** is still attached to the image forming apparatus **1**, that is, a direct replenishment system, is employed. Therefore, an operation of taking out the process cartridge **20** from the printer body **100** and replacing the process cartridge **20** by a brand-new process cartridge in the case where the amount of toner remaining in the process cartridge **20** has become small becomes unnecessary, and therefore the usability can be improved. To be noted, the image forming apparatus **1** and the toner pack **40** constitute an image forming system.
(92) To be noted, in the present embodiment, the reading apparatus **200** is provided in an upper portion of the image forming apparatus **1**, and in the case of opening the opening/closing member **83**, the reading apparatus **200** needs to be opened first to expose the top cover **82**. However, a configuration in which the reading apparatus **200** is omitted and the opening/closing member **83** is exposed in an upper portion of the image forming apparatus **1** from the beginning may be employed.
(93) (1-3) Reading Apparatus
(94) As illustrated in FIGS. **4**A and **4**B, the image reading apparatus **200** includes a reading unit

201 including an unillustrated reading portion therein, and a pressure plate **202** openably and closably supported by the reading unit **201**. A platen glass **203** that transmits light emitted from the reading portion and supports a document placed thereon is provided on the upper surface of the reading unit **201**.
(95) In the case of reading an image of a document by the reading apparatus **200**, the user places the document on the platen glass **203** in a state in which the pressure plate **202** is open. Then, the pressure plate **202** is closed to suppress displacement of the document on the platen glass **203**, and a reading command is output to the image forming apparatus **1** by, for example, operating the operation portion **300**. When the reading operation is started, the reading portion in the reading unit **201** reciprocates in a sub-scanning direction, that is, in the left-right direction in a state of facing the operation portion **300** of the image forming apparatus **1** on the front side. The reading portion receives light reflected on the document by a light receiving portion while radiating light onto the document from a light emitting portion, and reads the image of the document by performing photoelectric conversion.
(96) To be noted, in the description below, the front-rear direction, left-right direction, and up-down direction of the image forming apparatus **1** are defined on the basis of a state of facing the operation portion **300** on the front side as a standard. The up-down direction corresponds to the gravity direction. The positional relationship between members attachable to and detachable from the printer body **100** such as the process cartridge **20** will be described on the basis of a state where the members are attached to the printer body **100**. In addition, the “longitudinal direction” of the process cartridge **20** refers to an axial direction of the photosensitive drum **21**.
(97) (1-4) Configuration of Process Cartridge
(98) Next, a configuration of the process cartridge **20** will be described. FIG. **5**A is a perspective view of the process cartridge **20** and the toner pack **40**, and FIG. **5**B is a side view of the process cartridge **20** and the toner pack **40**. FIG. **6**A is a section view taken along a line **6**A-**6**A of FIG. **5**B, FIG. **6**B is a section view taken along a line **6**B-**6**B of FIG. **5**B, and FIG. **6**C is a section view taken along a line **6**C-**6**C of FIGS. **6**A and **6**B. To be noted, in FIGS. **5**A to **6**C, the outer shape of the replenishment container attaching portion **701** is illustrated in a simplified manner. For the detailed shape, see, for example, FIG. **9**A.
(99) As illustrated in FIGS. **5**A to **6**C, the process cartridge **20** is constituted by a toner receiving unit **801**, a developing unit **802**, and a cleaning unit **803**. The toner receiving unit **801**, the cleaning unit **803**, and the developing unit **802** are arranged in this order from the upper side to the lower side in the gravity direction. Each unit will be sequentially described below.
(100) The toner receiving unit **801** is disposed in an upper portion of the process cartridge **20**. A toner storage portion **8011** constituted by a frame member that stores toner is provided in the toner receiving unit **801**, and the replenishment container attaching portion **701** that couples to a toner pack **40** is provided at an end portion of the toner receiving unit **801**. To be noted, the frame member constituting the toner storage portion **8011** may be made up of a single member or a combination of a plurality of members. The replenishment container attaching portion **701** includes a replenishment port **8012** through which toner discharged from the toner pack **40** is received. The detailed configuration of the replenishment container attaching portion **701** and attachment of the toner pack **40** to the replenishment container attaching portion **701** will be described later.
(101) Further, a first conveyance member **8013**, a second conveyance member **8014**, and a third conveyance member **8015** are provided inside the toner receiving unit **801**. The first conveyance member **8013** conveys, in an arrow direction H illustrated in FIG. **6**C toward a center portion of the toner storage portion **8011**, toner that has fallen into an end portion of the toner storage portion **8011** in the longitudinal direction through the replenishment port **8012**. The second conveyance member **8014** conveys the toner conveyed by the first conveyance member **8013**, in an arrow J direction illustrated in FIG. **6**C perpendicular to the longitudinal direction, to an upper portion of the developing unit **802**, that is, to discharge ports **8016**. The third conveyance member **8015**

receives the toner from the second conveyance member **8014** mainly at a center portion in the longitudinal direction, and conveys the toner to a first side and a second side in the longitudinal direction, that is, in an arrow K direction and an arrow K′ direction. To be noted, the first to third conveyance members are operated so as to move the toner, and can be therefore also referred to as first to third developer moving members.
(102) When the toner from the toner pack **40** serving as a replenishment container flows into the toner receiving unit **801**, air also flows in. The replenishment container is also called a developer supply container. The toner receiving unit **801** includes an air filter **8017** illustrated in FIG. **5**A for allowing the air to flow in the arrow H direction when replenishing toner, such that it is easier to replenish toner. This air filter **8017** suppresses blowout of the toner from the replenishment port **8012** occurring as a result of the inner pressure of the toner receiving unit **801** increasing when replenishing toner and part of the air flowing in a direction opposite to the arrow H direction.
(103) Further, the discharge ports **8016** illustrated in FIG. **6**B for discharging toner from the toner storage portion **8011** to the developer container **32** of the developing unit **802** are respectively provided at two end portions of the toner receiving unit **801** in the longitudinal direction. The toner having reached the discharge ports **8016** by being conveyed by the third conveyance member **8015** falls into the developer container **32** in accordance with the gravity. To be noted, a conveyance member may be further provided in paths of the discharge ports **8016** to help the toner movement in accordance with the gravity.
(104) The developing unit **802** positioned in a lower portion of the process cartridge **20** includes openings **8021** illustrated in FIG. **6**B that receive the toner discharged through the discharge ports **8016**. Unillustrated sealing members are provided between the discharge ports **8016** and the openings **8021** such that the toner does not leak through a gap between the discharge ports **8016** and the openings **8021**.
(105) The toner having fallen into the toner receiving unit **801** from the toner pack **40** through the replenishment port **8012** is conveyed in the toner receiving unit **801** by the first conveyance member **8013**, the second conveyance member **8014**, and the third conveyance member **8015**. Then, the toner is delivered from the toner receiving unit **801** to the developing unit **802** through the discharge ports **8016** and openings **8021** provided at the two end portions in the longitudinal direction. In this manner, the toner supplied through the replenishment port **8012**, which is positioned at an end portion of the process cartridge **20** in the longitudinal direction and away from the developer container **32** in the horizontal direction as viewed in the longitudinal direction, is conveyed in the process cartridge **20** and reaches the developer container **3012**.
(106) As described above, the toner storage portion **8011** of the toner receiving unit **801** and the developer container **32** of the developing unit **802** communicate with each other, and thus constitute a storage container defining a space to store the toner in the process cartridge **20**. Therefore, in the present embodiment, the replenishment port **8012** for replenishing toner from the outside is provided as a part of the storage container of the process cartridge **20**. However, a replenishment port directly connected to the replenishment container may be provided in the printer body, and the process cartridge may receive the toner through this replenishment port. In this case, a part of the process cartridge **20** excluding the replenishment port is detachable from the image forming apparatus **1** as illustrated in FIG. **3**.
(107) The toner supplied to the developing unit **802** through the openings **8021** is stored in a conveyance chamber **36** formed in the developer container **32** constituted by a frame member of the developing unit **802** as illustrated in FIGS. **6**A and **6**B. To be noted, the frame member constituting the developer container **32** may be constituted by a single member or a combination of a plurality of members. Here, an agitation member **34** is provided in the conveyance chamber **36**. The agitation member **34** includes a shaft member **34***a* provided near the rotation center of the agitation member **34**, and a blade portion **34***b* extending in the radial direction from the shaft member **34***a*. In section view, toner within the rotation trajectory of the distal end of the blade

portion **34***b* is pushed and moved in accordance with the movement of the blade portion **34***b*. The toner replenished through the openings **8021** is conveyed toward the developing roller **31**, the supply roller **33**, and the developing blade **35** while being agitated by the agitation member **34**.

(108) The cleaning unit **803** includes a fourth conveyance member **8031**, a fifth conveyance member **8032**, and a waste toner chamber **8033** constituted by a frame member as illustrated in FIGS. **6**A and **6**B. To be noted, the frame member constituting the waste toner chamber **8033** may be made up of a single member or a combination of a plurality of members. The waste toner chamber **8033** is a space for storing collected matter, that is, so-called waste toner, such as transfer residual toner collected from the photosensitive drum **21** by the cleaning blade **24**, and is independent from the inner spaces of the toner receiving unit **801** and the developing unit **802**. The waste toner collected by the cleaning blade **24** is conveyed in an arrow M direction by the fourth conveyance member **8031** and the fifth conveyance member **8032**, and is gradually accumulated starting from the front side of a rear portion **8033***a* of the waste toner chamber **8033**.

(109) Here, a laser passing space SP that is a gap which the laser light L emitted from the scanner unit **11** illustrated in FIG. **1**A toward the photosensitive drum **21** can pass through is defined between the cleaning unit **803** and the developing unit **802** as illustrated in FIG. **6**A. As described above, the discharge ports **8016** and the openings **8021** for delivering the toner from the toner receiving unit **801** to the developing unit **802** are provided at end portions of the respective units in the longitudinal direction. Therefore, toner replenished from the outside of the image forming apparatus **1**, particularly through the replenishment port **8012** opening in the upper surface of the apparatus, can be conveyed to the developer container **32** provided in a lower portion of the process cartridge **20** while securing the laser passing space SP in a configuration of a small size as the whole of the process cartridge **20**.

(110) (1-5) Configuration of Toner Pack

(111) The configuration of the toner pack **40** will be described. FIG. **7**A is a perspective view of the toner pack **40** in a state in which a shutter member **41** is closed, and FIG. **7**B is a bottom view thereof. FIG. **8**A is a perspective view of the toner pack **40** in a state in which the shutter member **41** is open, FIG. **8**B is a bottom view thereof, and FIG. **8**C illustrates how the user squeezes the toner pack **40** with hands when replenishing toner. In addition, FIG. **12** is a perspective view of the toner pack **40** in the state in which the shutter member **41** is closed as viewed from below.

(112) As illustrated in FIGS. **7**A to **8**C, the toner pack **40** serving as an example of a replenishment container includes a bag member **43** filled with toner, a discharge portion **42** formed from resin and attached to the bag member **43**, and the shutter member **41** capable of opening and closing an opening portion of the discharge portion **42**. A memory unit **45** serving as a storage portion that stores information of the toner pack **40** is attached to the discharge portion **42**. The memory unit **45** includes, as a contact portion **45***a* that comes into contact with a contact portion **70133** of the replenishment container attaching portion **701** that is illustrated in FIGS. **9**A and **9**B and will be described later, a plurality of metal plates serving as metal terminals exposed to the outside of the toner pack **40**. In addition, as a material of the bag member **43**, polypropylene resin, polyethylene terephthalate resin, cardboards, paper, and so forth can be employed. In addition, the thickness of the bag member **43** can be set to 0.01 mm to 1.2 mm. In addition, the thickness is further preferably 0.05 mm to 1.0 mm from the viewpoint of squeezability for the user and the durability of the bag.

(113) As illustrated in FIGS. **7**B, **8**B, and **12**, the shutter member **41** has a shape obtained by cutting out a part of a disk relatively rotatable with respect to the discharge portion **42**. A side surface of the shutter member **41** extending in a thickness direction at the cutout portion functions as an engagement surface **41***s*. Meanwhile, the discharge portion **42** also has a shape having a cutout portion therein. The cutout portion of the discharge portion **42** includes an engagement surface **42***s* parallel to the engagement surface **41***s*. Further, a discharge port **42***a* is provided at a position at approximately 180° from the engagement surface **42***s* in the circumferential direction of the discharge port **42***a*. To be noted, details of the engagement surface **41***s* and **42***s* are illustrated in

FIG. **12**.

(114) As illustrated in FIGS. **7**B and **12**, when the positions of the cutouts of the shutter member **41** and the discharge portion **42** as viewed from above or below are aligned, the discharge port **42***a* is covered by the shutter member **41**. This state will be referred to as a closed state. As illustrated in FIG. **8**B, when the shutter member **41** rotates by 180° with respect to the discharge portion **42**, the discharge port **42***a* is exposed through the cutout portion of the shutter member **41**, and the inner space of the bag member **43** communicates with a space outside the toner pack **40**. To be noted, as illustrated in FIG. **12**, the shutter member **41** preferably has a structure in which a sealing layer **41***b* formed from an elastic material such as a sponge is stuck on a body portion **41***a* having stiffness. In this case, the sealing layer **41***b* is in firm contact with a sealing layer **42***c* covering a peripheral edge portion of the discharge port **42***a* in the closed state, and thus toner leakage is suppressed. The sealing layer **42***c* is illustrated in FIG. **12**, and is formed from an elastic material such as a sponge similarly to the sealing layer **41***b*.

(115) As will be described later, when replenishing the image forming apparatus **1** with toner from the toner pack **40**, the toner pack **40** is inserted in and coupled to the replenishment container attaching portion **701** by aligning the discharge portion **42** with a predetermined position. Then, when the discharge portion **42** is rotated by 180°, the discharge portion **42** relatively rotates with respect to the shutter member **41** to open the discharge port **42***a*, and the toner in the bag member **43** falls into the toner receiving unit **801** in accordance with the gravity. At this time, the shutter member **41** does not relatively move with respect to the replenishment container attaching portion **701**.

(116) As illustrated in FIG. **8**C, the user squeezes the bag member **43** in the state in which the toner pack **40** is attached to the replenishment container attaching portion **701** and rotated by 180°, and thus can promote discharge of toner from the toner pack **40**.

(117) To be noted, although the shutter member **41** that is rotatable has been described as an example herein, the shutter member may be omitted, and a shutter member of a slide type may be used instead of the rotary shutter member **41**. In addition, the shutter member **41** may be configured to be broken by attaching the toner pack **40** to a replenishment port **8012** or rotating the toner pack **40** in an attached state, or may have a detachable lid structure such as a sticker.

(118) In addition, it is preferable that a protective cap is attached to the discharge portion **42** of an unused toner pack **40** such that toner does not leak during transport or the like. For example, the protective cap engages with the cutout portions of the shutter member **41** and the discharge portion **42** in a state of being attached to the discharge portion **42** so as to restrict relative rotation of the shutter member **41** and the discharge portion **42**. By removing the protective cap, it becomes possible for the user to attach the toner pack **40** to the replenishment container attaching portion **701**.

(119) (1-6) Configuration of Replenishment Container Attaching Portion

(120) A shutter opening/closing mechanism of the toner pack **40** and the toner receiving unit **801**, and a locking mechanism of the shutter member **41** will be described. FIG. **9**A is a perspective view of the replenishment container attaching portion **701**, and FIG. **9**B is a top view of the replenishment container attaching portion **701**. The replenishment container attaching portion **701** includes the replenishment port **8012**, a replenishment port shutter **7013**, a locking member **7014**, and a rotation detection portion **7015**.

(121) The replenishment port **8012** is an opening portion communicating with the toner storage portion **8011** of the toner receiving unit **801** illustrated in FIG. **6**, and is fixed to the frame member **8010** of the toner receiving unit **801**. The replenishment port shutter **7013** includes a lid portion **70131** covering the replenishment port **8012**, a cylindrical portion **70132** that receives the discharge portion **42** of the toner pack **40**, and the contact portion **70133** connected to the contact portion **45***a* of the memory unit **45** of the toner pack **40** illustrated in FIG. **8**B. In FIG. **9**A, a part of the cylindrical portion **70132** covering the contact portion **70133** is indicated as a cylindrical portion

70132*a*. The replenishment port shutter **7013** is a member in which the lid portion **70131**, the cylindrical portion **70132**, and the contact portion **70133** are integrated, and is rotatably attached to the frame member **8010** of the toner receiving unit **801**. Each conductor exposed on the contact portion **70133** is electrically connected to a controller of the image forming apparatus **1** incorporated in the printer body **100**, via wiring provided in the process cartridge **20** and contacts between the process cartridge **20** and the printer body **100**.
(122) The rotation detection portion **7015** serving as a rotation detection sensor is a mechanism that detects the rotation of the replenishment port shutter **7013**. The rotation detection portion **7015** of the present embodiment is constituted by two conductive leaf springs **70151** and **70152**. The leaf spring **70152** springs in a clockwise direction, and when pressed by a projection portion **70135***a* provided on an outer periphery of the replenishment port shutter **7013**, comes into contact with the leaf spring **70151** at a distal end portion **701521**. That is, the rotation detection portion **7015** is an electric circuit configured such that a connected state and disconnected state thereof switch in accordance with the rotation angle, that is, rotational position of the replenishment port shutter **7013**. As will be described later, a controller **90** of the image forming apparatus **1** illustrated in FIG. **19** recognizes whether or not the discharge port **42***a* of the toner pack **40** communicates with the replenishment port **8012** of the replenishment container attaching portion **701**, on the basis of whether the rotation detection portion **7015** is in the connected state or the disconnected state. In other words, the controller **90** can determine that the replenishment operation by the user using the toner pack **40** has been normally performed at least up to the communication between the discharge port **42***a* and the replenishment port **8012**.
(123) A plurality of projection portions **70135***a* and **70135***b* are provided at an outer peripheral portion of the cylindrical portion **70132** of the replenishment port shutter **7013**. In addition, a plurality of projection portions **70125***a* and **70125***b* are also provided on a part of the frame member **8010** supporting the cylindrical portion **70132** of the replenishment port shutter **7013**, that is, a cylindrical portion **7011***a* of a portion **7011**. The plurality of projection portions **70125***a* and **70125***b* are positioned below the projection portion **70135***a* illustrated on the right side in FIG. **10**A in the gravity direction. The projection portion **70125***b* allows the projection portion **70135***a* illustrated on the right side in FIG. **10**A to pass through by rotational movement. In contrast, the projection portion **70135***a* illustrated on the left side in FIG. **10**A is positioned at the same height as the projection portion **70135***a* illustrated on the right side of FIG. **10**A, and extends downward to such a height as to overlap with the projection portions **70125***a* and **70125***b*. Therefore, the projection portion **70125***b* comes into contact with the projection portion **70135***a* illustrated on the left side in FIG. **10**A depending on the rotation angle, that is, rotational position of the replenishment port shutter **7013**, and thus restricts rotational movement of the projection portion **70135***a* illustrated on the left side in FIG. **10**A.
(124) In addition, before the replenishment port shutter **7013** rotates in an R1 direction, the projection portion **70125***a* comes into contact with the projection portion **70135***a* illustrated on the left side, and restricts the rotational movement of the projection portion **70135***a* in an R2 direction. In addition, the projection portion **70135***a* illustrated on the right side in FIG. **10**A abuts the locking member **7014**, and thus the rotational movement of the locking member **7014** in the R1 direction is restricted. In addition, after the replenishment port shutter **7013** has rotated in the R1 direction, the projection portion **70135***b* abuts the locking member **7014** that has moved to a locking position, and thus restricts the rotational movement of the locking member **7014** in the R2 direction. In addition, the projection portion **70135***a* illustrated on the right side in FIG. **10**A abuts the projection portion **70125***b*, and thus restricts further rotational movement of the projection portion **70135***a* in the R1 direction. To be noted, the rotation direction of the replenishment port shutter **7013** is the R1 direction when attaching the toner pack **40**, and is the R2 direction when detaching the toner pack **40**.
(125) The locking member **7014** is a member that restricts the rotation of the replenishment port

shutter **7013**. FIG. **11**A illustrates a state in which the locking member **7014** is in the locking position, and FIG. **11**B illustrates a state in which the locking member **7014** is in a lock releasing position. The locking member **7014** can be switched between the locking position serving as a restricting position and the lock releasing position serving as an allowing position by moving in the up-down direction. As illustrated in FIGS. **9**B and **11**A, when the locking member **7014** abuts the projection portion **70135***a* of the replenishment port shutter **7013** in the locking position, the rotation of the replenishment port shutter **7013** is restricted. When the locking member **7014** moves to the lock releasing position as illustrated in FIG. **11**B, the locking member **7014** retracts from the movement trajectory of the projection portion **70135***a* drawn when the replenishment port shutter **7013** moves, and thus the rotation of the replenishment port shutter **7013** is allowed.
(126) (1-7) Pressing Mechanism of Locking Member
(127) FIG. **13** illustrates a pressing mechanism **600** that moves the locking member **7014** between the locking position and the lock releasing position. The pressing mechanism **600** includes a motor **601**, an input gear **602**, a cam gear **603**, and an advancing/retracting pin **604**. The input gear **602** is a crossed helical gear attached to an output shaft of the motor **601**. The cam gear **603** includes a gear portion **6032** constituted by a helical gear that engages with the input gear **602**, and a cam portion **6031** for reciprocating the advancing/retracting pin **604**.
(128) The advancing/retracting pin **604** is supported by a holding member so as to be linearly movable in the gravity direction and an opposite direction thereto in the vertical direction. When the motor **601** rotates, the cam gear **603** is rotated via the input gear **602**, the advancing/retracting pin **604** reciprocates in the up-down direction by being pressed by the cam portion **6031**, and in accordance with this, the locking member **7014** also moves up and down between the locking position and the lock releasing position. FIG. **13** illustrates a locked state.
(129) To be noted, although a combination of a helical gear and a crossed helical gear has been used as the drive transmission configuration of the pressing mechanism **600** of the present embodiment, the configuration is not limited to this as long as the rotation of the motor can be converted into a linear motion. For example, a bevel gear may be used, or the input gear **602** may be removed and the cam gear **603** may be directly driven by the motor **601**. In addition, an actuator that outputs a linear motion such as a solenoid may be used as the drive source instead of the motor **601**.
(130) In addition, each member constituting the pressing mechanism **600** illustrated in FIG. **13** is supported by a frame member **609** of the printer body **100**. Meanwhile, a pivot shaft **7014***a* of the locking member **7014** is held by a holding portion provided on the frame member **8010** of the toner receiving unit **801** so as to be pivotable and slidable in the vertical direction. Therefore, when replacing the process cartridge **20**, the locking member **7014** is also replaced, and the pressing mechanism **600** is left in the printer body **100**. The pivot shaft **7014***a* and the advancing/retracting pin **604** are formed as separate members. When the locking member **7014** is in the lock releasing position, the advancing/retracting pin **604** is away from the locking member **7014**, and the process cartridge **20** is detached from the body with the advancing/retracting pin **604** left in the body. However, the configuration is not limited to this, and for example, the pivot shaft **7014***a* of the locking member **7014** may be supported by the printer body **100**.
(131) (1-8) Procedure of Replenishment Operation Using Toner Pack
(132) A procedure of the operation performed when detaching the toner pack **40** after attaching the toner pack **40** to the replenishment container attaching portion **701** and replenishing toner will be described on the basis of the configuration of the toner pack **40**, the replenishment container attaching portion **701**, and the pressing mechanism **600** described above. FIG. **10**A is a top view of the replenishment container attaching portion **701** when the replenishment port **8012** is in the closed state, and FIG. **10**B is a top view of the replenishment container attaching portion **701** when the replenishment port **8012** is in the open state.
(133) As illustrated in FIG. **10**A, the replenishment port shutter **7013** in the closed state is fixed so

as to be unrotatable with respect to the replenishment port **8012** by the projection portion **70135***a* abutting the locking member **7014** positioned in the locking position in the rotation direction. At this time, the lid portion **70131** of the replenishment port shutter **7013** completely blocks the replenishment port **8012**. In addition, the leaf springs **70151** and **70152** of the rotation detection portion **7015** are separated from each other, and the rotation detection portion **7015** is in the disconnected state.
(134) When inserting the toner pack **40** in the replenishment container attaching portion **701**, the user aligns the cutout portions of the discharge portion **42** of the toner pack **40** and the shutter member **41** illustrated in FIG. **12** with the replenishment port **8012** and the lid portion **70131** of the replenishment port shutter **7013** and inserts the toner pack **40**. In this case, the engagement surface **42***s* of the discharge portion **42** engages with an engagement surface **7013***s* illustrated in FIG. **9**C, which is a side surface of the lid portion **70131**, and the engagement surface **41***s* of the shutter member **41** engages with an engagement surface **8012***s* illustrated in FIG. **9**C, which is provided on an outer peripheral portion of the replenishment port **8012**. At this time, the discharge portion **42** engaging with the lid portion **70131** of the replenishment port shutter **7013** is unrotatable until the lock of the replenishment port shutter **7013** by the locking member **7014** is released later, and becomes rotatable together with the replenishment port shutter **7013** after the release of the lock. In addition, the shutter member **41** of the toner pack **40** is in an unrotatable state by engaging with the replenishment port **8012** fixed to the frame member **8010** of the toner receiving unit **801**. To be noted, as a different engagement mechanism of the lid portion **70131** and the discharge portion **42**, a projection portion projecting upward may be provided on the upper surface of the lid portion **70131** and a recess portion that engages with this projection portion may be provided on a lower surface **42***b* of the discharge portion **42** illustrated in FIG. **12**.
(135) In addition, by inserting the toner pack **40**, the contact portion **45***a* of the memory unit **45** illustrated in FIGS. **7**A and **7**B comes into contact with the contact portion **70133** of the replenishment container attaching portion **701**, and information stored in the memory unit **45** is read by the controller **90** of the image forming apparatus **1**. The memory unit **45** stores information indicating whether or not toner is in the toner pack **40**, that is, whether or not the toner pack **40** has been already used. This information will be also referred to as a brand-new product flag. When the controller **90** reads the brand-new product flag and determines that the toner pack **40** currently attached includes toner, that is, the toner pack **40** currently attached has not been used, the controller **90** controls the pressing mechanism **600** to push up the locking member **7014**. As a result of this, the locking member **7014** moves from the locking position to the lock releasing position illustrated in FIG. **11**B.
(136) In the state in which the locking member **7014** has moved to the lock releasing position, the locking member **7014** is separated from the projection portion **70135***a* of the replenishment port shutter **7013**, and thus the replenishment port shutter **7013** becomes rotatable in the R1 direction of FIGS. **10**A and **10**B. However, since the projection portion **70125***a* provided on the frame member **8010** of the toner receiving unit **801** interferes with the projection portion **70135***a* illustrated in FIG. **10**A, rotation of the replenishment port shutter **7013** in the R2 direction is restricted. That is, in FIG. **10**A, the projection portions **70125***a* and **70125***b* are positioned below the projection portions **70135***a* and **70135***b* such that the projection portions **70135***a* and **70135***b* can move and pass the projection portions **70125***a* and **70125***b* in the rotation direction.
(137) When the user grabs the toner pack **40** and rotates the discharge portion **42** or a portion of the bag member **43** close to the discharge portion **42** by 180° in the R1 direction, a state illustrated in FIG. **10**B is taken. The replenishment port shutter **7013** also rotates by 180° together with the discharge portion **42** of the toner pack **40**, thus the lid portion **70131** moves from the position covering the replenishment port **8012**, and the replenishment port **8012** is exposed. The side surface of the lid portion **70131** is pushed by the engagement surface **42***s*, which is a part of the discharge portion **42** that is rotating, and thus the lid portion **70131** rotationally moves together with the

engagement surface **42***s*. In addition, as a result of the discharge portion **42** rotating by 180° in a state in which the shutter member **41** is fixed, the discharge port **42***a* of the toner pack **40** illustrated in FIG. **8**B is exposed, and faces the replenishment port **8012**. As a result of this, the inner space of the toner pack **40** and the inner space of the toner receiving unit **801** communicate with each other through the discharge port **42***a* and the replenishment port **8012**, and the toner stored in the bag member **43** flows down into the toner storage portion **8011**.
(138) The toner having fallen into the toner storage portion **8011** is, as described above, conveyed inside the toner receiving unit **801**, reaches the developer container **32**, and becomes available for a developing process. To be noted, a configuration in which the developing unit **802** can perform the developing process as long as toner of an amount required for maintaining the image quality remains in the developer container **32** even before the newly replenished toner reaches the developer container **32** may be employed. That is, a configuration in which toner can be supplied to the developer container from a replenishment container disposed outside the image forming apparatus regardless of whether or not the image forming operation by the image forming portion **10** illustrated in FIG. **1**A is being performed may be employed.
(139) In addition, the projection portion **70125***b* is disposed so as to abut the projection portion **70135***a* of the replenishment port shutter **7013** when the replenishment port shutter **7013** is rotated by 180° in the R1 direction from the state of FIG. **10**A as illustrated in FIG. **10**B. That is, the projection portion **70125***b* is also positioned below the projection portions **70135***a* and **70135***b* similarly to the projection portion **70125***a*. As a result of this, pivoting of the replenishment port shutter **7013** beyond 180° in the R1 direction is restricted. At the same time, the projection portion **70135***a* of the replenishment port shutter **7013** presses the leaf spring **70152** of the rotation detection portion **7015**, and the distal end portion **701521** thereof is brought into contact with the leaf spring **70151**. When the rotation detection portion **7015** is in the connected state, the controller **90** recognizes that the replenishment port shutter **7013** has transitioned to the open state, and operates the pressing mechanism **600** to move the locking member **7014** again to the locking position. Then, the locking member **7014** engages with the projection portion **70135***b* of the replenishment port shutter **7013** to restrict the rotation in the R2 direction, and thus the replenishment port shutter **7013** and the toner pack **40** both become unrotatable in any direction.
(140) Further, in the state of FIG. **10**B in which the discharge portion **42** of the toner pack **40** and the replenishment port shutter **7013** have been rotated by 180°, the lid portion **70131** of the replenishment port shutter **7013** covers an upper portion of the shutter member **41** of the toner pack **40**. Therefore, when it is attempted to pick up the toner pack **40** from the replenishment container attaching portion **701**, the shutter member **41** interferes with the lid portion **70131**, and the movement of the toner pack **40** is restricted. Therefore, detachment of the toner pack **40** from the replenishment container attaching portion **701** is suppressed unless the user performs the detachment operation of the toner pack **40** in accordance with a predetermined procedure that will be described below.
(141) After the start of discharge of toner from the toner pack **40**, if a condition for determining that the discharge of toner has been completed is satisfied, the controller **90** operates the pressing mechanism **600** to move the locking member **7014** to the lock releasing position. In the present embodiment, completion of the discharge of toner is determined on the basis of the time elapsed from the time point at which the rotation detection portion **7015** has transitioned to the connected state.
(142) After the locking member **7014** has moved to the lock releasing position, the user can detach the toner pack **40** by following a procedure reversed from the procedure performed when attaching the toner pack **40**. That is, the user grabs the discharge portion **42** of the toner pack **40** or a part of the bag member **43** close to the discharge portion **42**, and rotates the toner pack **40** by 180° in the R2 direction, which is opposite to the direction of rotation at the time of attachment. In this case, the replenishment port shutter **7013** rotates by 180° together with the discharge portion **42**, and the

replenishment port **8012** is covered by the lid portion **70131** of the replenishment port shutter **7013** as illustrated in FIG. **10**A. In addition, the projection portion **70135***a* of the replenishment port shutter **7013** illustrated on the left side in FIG. **10**A abuts the projection portion **70125***a*, and thus the rotation of the replenishment port shutter **7013** beyond 180° in the R2 direction is restricted.
(143) In the state in which the discharge portion **42** of the toner pack **40** has been rotated by 180° in the R2 direction, the position of the cutout portion of the discharge portion **42** and the position of the cutout portion of the shutter member **41** are aligned as illustrated in FIG. **12**. Therefore, even if the toner pack **40** is moved upward, the shutter member **41** does not interfere with the lid portion **70131** of the replenishment port shutter **7013**, and therefore the user can detach the toner pack **40** from the replenishment container attaching portion **701** by grabbing and lifting the toner pack **40**.
(144) To be noted, in the course of rotating the replenishment port shutter **7013** by 180° in the R2 direction, the projection portion **70135***a* is separated from the leaf spring **70152**, and the rotation detection portion **7015** returns to the disconnected state. Then, the controller **90** recognizes that the replenishment port shutter **7013** has transitioned to the closed state, and operates the pressing mechanism **600** to move the locking member **7014** to the locking position. As a result of this, the replenishment container attaching portion **701** transitions back to the initial state as before the toner replenishment operation is performed. For example, the controller **90** may determine that a predetermined condition to move the locking member **7014** to the lock releasing position is satisfied when a predetermined time has elapsed after the rotation detection portion **7015** has transitioned to the connected state. To be noted, the trigger for moving the locking member **7014** to the locking position may be loss of connection between the contact portion **70133** of the replenishment container attaching portion **701** and the contact portion **45***a* of the toner pack **40** illustrated in FIG. **7** caused by detachment of the toner pack **40** from the replenishment container attaching portion **701**.
(145) Although the positional relationship is set such that the discharge port **42***a* of the toner pack **40** and the replenishment port **8012** communicate with each other after the rotation by 180° in the present embodiment, the rotation angle required for the communication may be changed as long as the detachment of the toner pack **40** is made possible by an operation similar to that of the present embodiment.
(146) (1-9) Panel
(147) Next, a panel **400** will be described. For example, the Panel **400** is provided on the front surface of the casing of the printer body **100** as illustrated in FIGS. **1**B and **14**A to **14**C. The panel **400** is an example of a display portion that displays information related to the remainder amount of toner in the developer container **32**, or a remaining capacity of the developer container **32**. The panel **400** is constituted by a liquid crystal panel including a plurality of indicators. In the present embodiment, three indicators **4001**, **4002**, and **4003** are arranged in this order from the upper side to the lower side in the vertical direction. The panel **400** indicates the amount of toner that can be added to the developer container **32** for replenishment by the display of the indicators **4001** to **4003** that changes stepwise. The controller **90** constantly updates the display of the panel **400** on the basis of replenishment operation completion recognition that will be described later. In addition, in the case where the completion of the replenishment operation is not reflected on the toner remainder amount, the toner remainder amount may be detected subsequently, and the display of the panel **400** may be updated. For example, in the case where the controller **90** has detected by an optical sensor denoted by **51***a* and **51***b* that actually the toner has not been sufficiently replenished after the light of the indicator **4002** has been turned on, the controller **90** updates the display of the panel **400** by turning off the light of the indicator **4002**. In addition, the lowermost indicator **4003** also indicates whether the toner in the developer container **32** is at a Low level or at an Out level. To be noted, the Low level is a level at which, although the developer container **32** needs to be replenished with toner, at least toner of an amount required for maintaining the image quality remains and the image forming operation can be still performed. The Out level is a level at which

almost no toner remains in the developer container **32** and the image forming operation cannot be performed.
(148) In the illustrated configuration example of the panel **400**, lights of the three indicators **4001** to **4003** all being off indicates that the toner in the developer container **32** is at the Out level. This state serves as a fourth state.
(149) In the case where only the light of the lower indicator **4003** is on as illustrated in FIG. **14**A, the toner remainder amount in the developer container **32** is at the Low level. In this state, lights of two of the indicators are off, and therefore it can be seen that toner of an amount corresponding to two toner packs **40** can be added for replenishment. This state serves as a third state. In addition, it can be also seen that toner of an amount corresponding to two toner packs **40** can be added for replenishment from the fact that lights of number panels “+1” and “+2” next to the indicators are on.
(150) In the case where lights of the middle and lower indicators **4002** and **4003** are on and the light of the upper indicator **4001** is off as illustrated in FIG. **14**B, the toner remainder amount in the developer container **32** is larger than that of the Low level and smaller than that of a Full level in which the developer container **32** is full. In this state, the light of one indicator is off, and therefore it can be seen that, for example, toner of an amount corresponding to one toner pack **40** can be added for replenishment. This state serves as a second state. In addition, it can be also seen that toner of an amount corresponding to one toner pack **40** can be added for replenishment from the fact that the light of the number panel “+1” next to an indicator is on and the light of the number panel “+2” next to an indicator is off.
(151) In the case where all the three indicators **4001** to **4003** are on as illustrated in FIG. **14**C, the toner remainder amount in the developer container **32** is at the Full level. In this state, light of no indicator is off, and therefore it can be seen that, for example, no toner can be added for replenishment from the toner pack **40**. This state serves as a first state. In addition, it can be also seen that no toner can be added for replenishment from the toner pack **40** from the fact that the lights of the number panels “+1” and “+2” next to the indicators are off.
(152) To be noted, the panel **400** illustrated in FIGS. **14**A to **14**C is an example of a display portion whose display content changes in accordance with the toner remainder amount in the developer container **32**, and a different configuration may be employed. For example, the panel may be constituted by a combination of a light source such as an LED or an incandescent lamp and a diffusion lens instead of a liquid crystal panel. Alternatively, a configuration in which the indicators are omitted and only the number panels are used or a configuration in which the number panels are omitted and only the indicators are used may be employed.
(153) In addition, the number and display method of the indicators of the panel **400** may be appropriately modified. For example, the user may be prompted to replenish toner by flickering the light of the lower indicator in the case where the toner remainder amount in the developer container **32** is at the Low level.
(2) First Modification Example
(154) Next, a first modification example in which a toner bottle unit having a bottle shape is used as another example of a replenishment container instead of the toner pack having a bag shape will be described with reference to FIGS. **15**A to **15**D. To be noted, this tonner bottle unit is configured to be attachable to and detachable from the replenishment container attaching portion **701** described above similarly to the toner pack **40** described above. Therefore, description of elements of the image forming apparatus that are the same as in the first embodiment will be omitted.
(155) (2-1) Configuration of Toner Bottle Unit
(156) FIG. **15**A is a perspective view of a toner bottle unit **900** illustrating the external appearance thereof, and FIG. **15**B is a perspective view of the toner bottle unit **900** after discharge of toner. FIG. **15**C is a diagram illustrating the toner bottle unit **900** as viewed from the lower side of a piston, and FIG. **15**D is a section view of the toner bottle unit **900** taken along a line D-D of FIG.

15C.

(157) In addition, FIG. **16**A is a perspective view of the toner bottle unit **900** in which illustration of an outer cylinder **903** illustrated in FIG. **15**A is omitted, and FIG. **16**B is a perspective view of the toner bottle unit **900** after the discharge of toner in which illustration of the outer cylinder **903** is omitted. FIG. **16**C is a diagram illustrating a state before a push-in operation of a component related to push-in detection of the toner bottle unit **900**, and FIG. **16**D is a diagram illustrating a state after the push-in operation of the component related to push-in detection. FIG. **16**E is a diagram illustrating a state before a rotating operation of a component related to rotation detection of the toner bottle unit **900**, and FIG. **16**F is a diagram illustrating a state after the rotating operation of the component related to the rotation detection of the toner bottle unit **900**.

(158) As illustrated in FIGS. **15**A and **15**D, the toner bottle unit **900** roughly includes the outer cylinder **903**, an inner cylinder **901**, a piston **902**, a shutter member **904**, and a memory unit **911**. The outer cylinder **903** and the inner cylinder **901** have cylindrical shapes, the inner cylinder **901** is fit inside the outer cylinder **903**, and the piston **902** is fit inside the inner cylinder **901** and is slidable with respect to the inner cylinder **901**. In the description below, the direction in which the piston **902** moves, that is, the direction of the axis of the outer cylinder **903** and the inner cylinder **901** will be referred to as the axial direction of the toner bottle unit **900**. In addition, the piston **902** serves as an example of a pressing member.

(159) The inner cylinder **901** includes a toner storage portion **9014** that has a cylindrical shape and stores toner, a bottom portion **9013** provided on a first end side in the axial direction, and a discharge port **9011** provided in the bottom portion **9013**. The inner cylinder **901** has a cylindrical shape in which a first end portion of the toner storage portion **9014** in the axial direction is closed by the bottom portion **9013**. An opening portion **9012** is provided on a second end side of the toner storage portion **9014**, and the piston **902** is inserted in the toner storage portion **9014** through the opening portion **9012**. In addition, a weight member **905** having a spherical shape and movable in the toner storage portion **9014** is included in the inner cylinder **901**.

(160) The outer cylinder **903** includes an inner cylinder accommodating portion **9034** having a cylindrical shape that accommodates the toner storage portion **9014** of the inner cylinder **901** therein, a bottom portion **9033** provided on the first end side in the axial direction, and a discharge port **9031** provided in the bottom portion **9033**. The outer cylinder **903** has a cylindrical shape in which a first end portion of the inner cylinder accommodating portion **9034** in the axial direction is closed by the bottom portion **9033** similarly to the inner cylinder **901**, and holds the inner cylinder **901** relatively unmovably. An opening portion **9032** through which the piston **902** is inserted is provided on the second end side of the inner cylinder accommodating portion **9034**.

(161) The discharge port **9011** of the inner cylinder **901** has a thin cylindrical shape extending from the bottom portion **9013** toward the first end side in the axial direction. The discharge port **9031** of the outer cylinder **903** is provided at a position corresponding to the discharge port **9011** of the inner cylinder **901** in the bottom portion **9033**. The discharge port **9031** of the outer cylinder **903** is a discharge port through which the toner stored in the toner storage portion **9014** is discharged to the outside of the toner bottle unit **900**. To be noted, a retracting space **9013***a* for the weight member **905** to retract into so as not to block the discharge port **9011** when pushing the piston **902** in is provided adjacent to the discharge port **9011** of the inner cylinder **901**.

(162) To be noted, the bottom portion **9013** of the inner cylinder **901** has an inclined shape whose sectional area is smaller on the discharge port side in the axial direction, particularly a conical shape whose inner diameter is smaller on the discharge port side in the axial direction. The bottom portion **9033** of the outer cylinder **903** opposing the bottom portion **9013** of the inner cylinder **901** also has a similar inclined shape. The discharge port **9011** of the inner cylinder **901** and the retracting space **9013***a* are provided at a vertex portion of the inclined shape of the bottom portion **9033**. The weight member **905** has a spherical shape, and is guided by the bottom portion **9013** to move to the retracting space **9013***a* by the gravity.

(163) The piston **902** includes an elastic member **906** attached to a first end portion **9023** on the first end side in the axial direction, that is, on the discharge port side, and a push-in rib **9021** provided in the vicinity of a second end portion **9022** on the second end side, which is a part that the user pushes when pushing in the piston **902**. The elastic member **906** is configured to come into contact with the inner circumferential surface of the toner storage portion **9014** with no gap therebetween, and has a function of suppressing leakage of toner when pushing in the piston **902**. In addition, the push-in rib **9021** is a projection shape projecting outward in the radial direction from the outer circumferential surface of the piston **902**.
(164) The configuration of the shutter member **904** is similar to that of the shutter member **41** provided in the toner pack **40** described above. That is, as illustrated in FIG. **15**C, the shutter member **904** has a shape of a disk partially cut out and relatively rotatable with respect to the outer cylinder **903**. A side surface of the shutter member **904** extending in the thickness direction in the cutout portion functions as an engagement surface **904***s*. Meanwhile, the outer cylinder **903** also has a shape with a cutout. The outer cylinder **903** includes an engagement surface **903***s* parallel to the engagement surface **904***s* in the cutout portion. In addition, the discharge port **9031** is provided at a position away from the engagement surface **903***s* by approximately 180° in the circumferential direction of the outer cylinder **903**.
(165) FIG. **15**C illustrates a state in which the discharge port **9031** is already exposed, but in the state at the time when the toner bottle unit **900** is shipped, the positions of the cutout engagement surfaces **903***s* and **904***s* of the shutter member **904** and the outer cylinder **903** are aligned. In this case, the discharge port **9031** is covered by the shutter member **904**, and the sealed state of the toner storage portion **9014**, that is, the closed state is maintained. As illustrated in FIG. **15**C, when the shutter member **904** is rotated by 180° with respect to the outer cylinder **903**, the discharge port **9031** is exposed through the cutout portion of the shutter member **904**, thus the sealing of the toner storage portion **9014** is cancelled, and it becomes possible to discharge the toner. This state corresponds to the open state. The configuration of the discharge port **9031**, the engagement surface **903***s*, and the shutter member **904** are basically the same as the configuration described with reference to FIGS. **7**A to **8**C and **12**.
(166) A memory unit **911** serving as a storage portion that stores information of the toner bottle unit **900** is attached to a portion near the discharge port **9031** of the outer cylinder **903**. The memory unit **911** includes a plurality of metal plates **9111**, **9112**, and **9113** illustrated in FIG. **16**A exposed to the outside of the toner bottle unit **900** as a contact portion **911***a* that comes into contact with the contact portion **70133** of the replenishment container attaching portion **701** illustrated in FIG. **9**A.
(167) (2-2) Push-In Detection Mechanism of Piston
(168) In addition, as illustrated in FIGS. **16**A and **16**C, as a push-in detection mechanism that detects a push-in operation of the piston **902**, a push-in detection rod **907**, a first contact plate **908**, and a second contact plate **909** are disposed between the outer cylinder **903** and the inner cylinder **901**. The push-in detection rod **907** is formed from an insulating material such as a resin, and the first contact plate **908** and the second contact plate **909** are formed from a conductive material such as metal. The push-in detection rod **907** includes a contact cancelling portion **9072** on the first end side in the axial direction, that is, on the discharge port side, and a piston contact portion **9071** capable of abutting the push-in rib **9021** of the piston **902** on the second end side in the axial direction. The push-in detection rod **907** moves in the axial direction in accordance with the push-in operation of the piston **902** as a result of the push-in rib **9021** pressing the piston contact portion **9071**.
(169) For example, the push-in detection rod **907** is fit in a groove shape defined in the axial direction in the outer circumferential surface of the inner cylinder **901** or the inner circumferential surface of the outer cylinder **903**, and is thus held so as to be movable in the axial direction with respect to the inner cylinder **901** and the outer cylinder **903** while the movement of the push-in detection rod **907** in a direction perpendicular to the axial direction is restricted. In addition, the

piston contact portion **9071** has a shape bent perpendicularly to the axial direction, that is, a shape bent into an L shape such that the push-in rib **9021** more reliably abuts the piston contact portion **9071**. To be noted, although the push-in rib **9021** is provided to extend all around the piston **902** on the outer circumferential surface of the piston **902** in FIG. **16**A, a configuration in which the push-in rib **9021** is formed in only a position corresponding to the piston contact portion **9071** in the circumferential direction may be employed.

(170) The first contact plate **908** and the second contact plate **909** are metal plates whose connected state and disconnected state are switched in accordance with the position of the push-in detection rod **907** formed from an insulating resin. A brand-new product detection method of the toner bottle unit **900** using the first contact plate **908** and the second contact plate **909** will be described later.

(171) In addition, a cylinder cover **910** illustrated in FIG. **15**A is provided at an end portion of the outer cylinder **903** on the opening portion side so as to suppress dropping of the push-in detection rod **907**. That is, the cylinder cover **910** defining the opening portion **9032** of the outer cylinder **903** is narrowed such that the edge of the opening portion **9032** is further on the inside than the outer edge of the piston contact portion **9071** illustrated in FIG. **16**B in the radial direction as illustrated in FIG. **15**D. Therefore, even when a force to move the push-in detection rod **907** toward the opening portion side in the axial direction is applied, the piston contact portion **9071** interferes with the cylinder cover **910**, and therefore the push-in detection rod **907** does not drop from the toner bottle unit **900**.

(172) (2-3) Brand-New/Used Determination of Toner Bottle Unit

(173) Next, a configuration for detecting whether the toner bottle unit **900** is unused, that is, brand-new, or used when attaching the toner bottle unit **900** to the replenishment container attaching portion **701** will be described. As illustrated in FIGS. **16**C and **16**D, the contact cancelling portion **9072** of the push-in detection rod **907** is positioned near the first contact plate **908** and the second contact plate **909**.

(174) FIG. **16**C corresponds to a state before the piston push-in illustrated in FIG. **16**A, and the first contact plate **908** and the second contact plate **909** are in contact with each other and thus are in the connected state. At this time, it is preferable that the one of the first contact plate **908** and the second contact plate **909** that are formed from metal is formed in a leaf spring shape and is in pressure contact with the other. In addition, for example, the conduction between the first contact plate **908** and the second contact plate **909** can be made more reliable by applying a conductive grease on the contact surfaces of the first contact plate **908** and the second contact plate **909**.

(175) FIG. **16**D corresponds to a state after the piston push-in illustrated in FIG. **16**B, and the first contact plate **908** and the second contact plate **909** are in the disconnected state. In this state, the contact cancelling portion **9072** of the push-in detection rod **907** pushed in by the push-in rib **9021** gets between the first contact plate **908** and the second contact plate **909**, and thus physically separate the first contact plate **908** and the second contact plate **909**. At least the contact cancelling portion **9072** of the push-in detection rod **907** is formed from an insulating material, and the conduction between the first contact plate **908** and the second contact plate **909** is disconnected in the state of FIG. **16**D in which the contact cancelling portion **9072** is present therebetween.

(176) The first contact plate **908** and the second contact plate **909** are connected to different metal plates among the plurality of metal plates **9111** to **9113**, at end portions opposite to end portions that come into contact with the contact cancelling portion **9072** of the push-in detection rod **907**. Here, the first contact plate **908** is connected to the metal plate **9111**, and the second contact plate **909** is connected to the metal plate **9113**. In this case, whether the toner bottle unit **900** is in a state before the piston push-in or in a state after the piston push-in, that is, whether the toner bottle unit **900** is unused or used can be determined by detecting whether a current is generated when a minute voltage is applied between the metal plates **9111** and **9113**. That is, in a state in which the toner bottle unit **900** is attached to the replenishment container attaching portion **701**, the controller **90** of the image forming apparatus **1** can determine whether the toner bottle unit **900** is used or unused,

on the basis of presence/absence of conduction between the metal plates **9111** and **9113**. In addition, the controller **90** can determine that the replenishment operation by the user has been finished, on the basis of disconnection between the first contact plate **908** and the second contact plate **909**. On the basis of this determination, the controller **90** performs display control of the panel **400** described above. In addition, the controller **90** writes, in the memory unit **45** and in accordance with the change in the conduction between the metal plates **9111** and **9113**, a brand-new product flag indicating whether or not the toner bottle unit **900** is used. The brand-new product flag being 1 corresponds to being brand-new, and the brand-new product flag being 0 corresponds to having been used.

(177) To be noted, in the case of the configuration described above, the memory unit **911** is preferably disposed in a circuit connecting the metal plates **9111** and **9112**. As a result of this, the controller **90** of the image forming apparatus can access the memory unit **911** through the metal plates **9111** and **9112** while monitoring the push-in operation of the toner bottle unit **900** via the metal plates **9111** and **9113**.

(178) (2-4) Rotation Detection of Toner Bottle Unit

(179) Next, a method for detecting the rotation of the toner bottle unit **900** will be described with reference to FIGS. **16**E and **16**F. To be noted, the rotation detection method of the present embodiment is the same as in the embodiment described above in which the toner pack **40** is used, except that the shutter member **904** that seals the discharge port of the replenishment container is attached to the outer cylinder **903** of the toner bottle unit **900**.

(180) As illustrated in FIGS. **16**E and **16**F, the two conductive leaf springs **70151** and **70152** are provided in the replenishment container attaching portion **701** of the process cartridge **20** as the rotation detection portion **7015**. In addition, the projection portion **70135***b* is provided on an outer peripheral portion of the replenishment port shutter **7013**.

(181) As illustrated in FIG. **16**E, in a state before the toner bottle unit **900** inserted in the replenishment container attaching portion **701** is rotated, the distal end portion **701521** of the leaf spring **70152** is not in contact with the leaf spring **70151**, and therefore the rotation detection portion **7015** is in the disconnected state. That is, no current flows when a minute voltage is applied between the leaf springs **70151** and **70152**. As illustrated in FIG. **16**F, when the toner bottle unit **900** is rotated by 180°, the leaf spring **70152** is pressed by the projection portion **70135***a*, thus the distal end portion **701521** comes into contact with the leaf spring **70151**, and the rotation detection portion **7015** is switched to the connected state. In this state, a current flows when a minute voltage is applied between the plate springs **70151** and **70152**. The controller **90** of the image forming apparatus **1** recognizes whether or not the discharge port **9031** of the toner bottle unit **900** and the replenishment port **8012** of the replenishment container attaching portion **701** communicate with each other, on the basis of whether the rotation detection portion **7015** is in the connected state or in the disconnected state.

(182) (2-5) Flow of Replenishment Operation Using Toner Bottle Unit

(183) A series of operation for detaching the toner bottle unit **900** after attaching the toner bottle unit **900** to the replenishment container attaching portion **701** and replenishing toner will be described. To be noted, description of elements same as in the embodiment described above where the toner pack **40** is used will be omitted.

(184) First, the user attaches an unused toner bottle unit **900** to the replenishment container attaching portion **701**. Specifically, the cutout engagement surfaces **903***s* and **904***s* of the outer cylinder **903** and the shutter member **904** illustrated in FIG. **15**C are aligned with the replenishment port **8012** and the lid portion **70131** of the replenishment port shutter **7013**, and the toner bottle unit **900** is inserted. In this case, the engagement surface **903***s* of the outer cylinder **903** engages with the engagement surface **7013***s*, which is a side surface of the lid portion **70131**, and the engagement surface **904***s* of the shutter member **904** engages with the engagement surface **8012***s* provided on an outer peripheral portion of the replenishment port **8012**. At this time, the outer cylinder **903**

engaging with the lid portion **70131** of the replenishment port shutter **7013** is unrotatable until the lock of the replenishment port shutter **7013** by the locking member **7014** is released later, and becomes rotatable together with the replenishment port shutter **7013** after the release of the lock. In addition, the shutter member **904** is in an unrotatable state by engaging with the replenishment port **8012** fixed to the frame member **8010** of the toner receiving unit **801**. Further, the leaf springs **70151** and **70152** of the rotation detection portion **7015** are away from each other, and the rotation detection portion **7015** is in the disconnected state as illustrated in FIG. **16**E.

(185) In the case where an unused toner bottle unit **900** is inserted in the replenishment container attaching portion **701**, the controller **90** recognizes that the toner bottle unit **900** is brand-new by the brand-new product detection mechanism described above. The controller **90** may recognize the conduction between the metal plates **9111** and **9113** or make a determination by reading the brand-new product flag in the memory unit **45**. The brand-new product flag being 1 corresponds to being brand-new, and the brand-new product flag being 0 corresponds to having been used. In this case, the controller **90** operates the pressing mechanism **600** to move the locking member **7014** to the lock releasing position, and thus the toner bottle unit **900** becomes rotatable.

(186) Then, when the user grabs the toner bottle unit **900** and rotates the toner bottle unit **900** by 180°, the shutter member **904** and the replenishment port shutter **7013** are opened, and the discharge port **9031** of the toner bottle unit **900** and the replenishment port **8012** of the replenishment container attaching portion **701** communicate with each other. The operation of opening the shutter member **904** and the replenishment port shutter **7013** in accordance with the rotation of the toner bottle unit **900** is similar to the case of the toner pack **40** described with reference to FIGS. **10**A and **10**B.

(187) As illustrated in FIG. **16**F, in a state in which the toner bottle unit **900** is rotated by 180°, the distal end portion **701521** of the leaf spring **70152** pressed by the projection portion **70135***b* of the replenishment port shutter **7013** comes into contact with the leaf spring **70151**. When the rotation detection portion **7015** is switched to the connected state in this manner, the controller **90** of the image forming apparatus **1** detects that the rotation operation of the toner bottle unit **900** has been performed. That is, the controller **90** recognizes that the sealing by the shutter member **904** and the replenishment port shutter **7013** has been cancelled and the discharge port **42***a* of the toner pack **40** and the replenishment port **8012** of the replenishment container attaching portion **701** communicate with each other. In addition, the controller **90** operates the pressing mechanism **600** to move the locking member **7014** to the locking position, and thus restricts the rotation of the toner bottle unit **900**.

(188) Next, the user presses the piston **902** of the toner bottle unit **900** to start discharge of toner. The toner having fallen into the toner storage portion **8011** is conveyed inside the toner receiving unit **801** and reaches the developer container **32**. Also in the present modification example, when the piston **902** is pushed to the deepest position, the push-in detection mechanism described above detects that the push-in operation of the piston **902** has been completed. That is, as illustrated in FIG. **16**B, the push-in rib **9021** of the piston **902** presses the piston contact portion **9071** of the push-in detection rod **907**, and thus the push-in detection rod **907** moves accompanied by the piston **902**.

(189) Then, as illustrated in FIG. **16**D, the contact cancelling portion **9072** of the push-in detection rod **907** disconnects the conduction between the first contact plate **908** and the second contact plate **909**. The controller **90** of the image forming apparatus **1** recognizes the completion of the push-in of the piston **902** on the basis of the fact that no longer a current flows even if a voltage is applied between the metal plate **9111** connected to the first contact plate **908** and the metal plate **9113** connected to the second contact plate **909**. That is, in the present modification example, detection of completion of the push-in operation of the piston **902** by the push-in detection mechanism serves as a condition for determining that discharge of toner is completed. To be noted, a configuration in which the controller **90** rewrites the brand-new product flag in the memory unit **911** in the case

where the conduction between the first contact plate **908** and the second contact plate **909** is disconnected, and determines that the discharge of toner has been completed on the basis of the rewriting of the brand-new flag may be employed.

(190) The controller **90** that has detected the completion of discharge of toner from the toner bottle unit **900** operates the pressing mechanism **600** again to move the locking member **7014** to the lock releasing position, and thus makes the toner bottle unit **900** rotatable. The user grabs the toner bottle unit **900** and rotates the toner bottle unit **900** by 180°. In this case, the discharge port **9031** of the toner bottle unit **900** is covered by the shutter member **904**, and the replenishment port **8012** of the replenishment container attaching portion **701** is covered by the lid portion **70131** of the replenishment port shutter **7013**. In addition, the leaf springs **70151** and **70152** are separated as illustrated in FIG. **16**E, and the rotation detection portion **7015** returns to the disconnected state. Then, the controller **90** recognizes that the replenishment port shutter **7013** has been switched to the closed state, and operates the pressing mechanism **600** to move the locking member **7014** to the locking position. As a result of this, the replenishment container attaching portion **701** returns to the initial state before the toner replenishment.

(3) Second Modification Example

(191) Next, a second modification example in which the configuration of the process cartridge is different will be described. The present modification example has the same elements as in the first embodiment except for elements related to the process cartridge, and therefore description of the same elements will be omitted.

(192) (3-1) Process Cartridge

(193) FIGS. **17**A to **17**D are respectively a perspective view, a side view, a section view, and another section view of a process cartridge **20**A according to the present modification example. FIGS. **17**C and **17**D are section views taken at cutting positions respectively illustrated in FIG. **17**B.

(194) As illustrated in FIGS. **17**A to **17**D, the process cartridge **20**A of the present modification example includes the toner receiving unit **801**, the developing unit **802**, and a drum unit **803**A. In contrast with the first embodiment, the drum unit **803**A does not include the cleaning blade **24** that cleans the surface of the photosensitive drum **21** or the waste toner chamber **8033** illustrated in FIG. **6**A that accommodates waste toner. This is because a cleanerless configuration is employed in the present modification example. In the cleanerless configuration, the transfer residual toner remaining on the surface of the photosensitive drum **21** without being transferred onto the recording material is collected into the developing unit **802** and reused is employed. To be noted, for example, nonmagnetic or magnetic one-component developer is also used herein.

(195) In the illustrated example, the developing unit **802** is positioned in a lower portion of the process cartridge **20**A, and the toner receiving unit **801** and the drum unit **803**A are positioned above the developing unit **802** in the gravity direction. Although the toner receiving unit **801** and the drum unit **803**A do not overlap as viewed in the gravity direction as illustrated in FIG. **17**B, the two may be aligned in the up-down direction at least partially. In addition, the toner receiving unit **801** is disposed in the space where the cleaning blade **24** and the waste toner chamber **8033** are provided in the first embodiment. The configuration of the replenishment container attaching portion **701** provided in the toner receiving unit **801** is the same as in the first embodiment, and FIGS. **17**A to **17**D illustrate a simplified shape thereof.

(196) A laser passing space SP serving as a gap for the laser light L emitted from the scanner unit **11** illustrated in FIG. **1**A toward the photosensitive drum **21** to pass through is defined between the developing unit **802**, the drum unit **803**A, and the toner receiving unit **801**. In addition, it is preferable that, in the drum unit **803**A, a pre-exposing unit for removing the electrostatic latent image by radiating light onto the surface of the photosensitive drum **21** is disposed downstream of the transfer portion and between the transfer portion and the charging roller **22** in the rotation direction of the photosensitive drum **21**.

(197) (3-2) Behavior of Toner in Cleanerless Configuration
(198) The behavior of toner in the cleanerless configuration will be described. The transfer residual toner remaining on the photosensitive drum **21** in the transfer portion is removed in accordance with the following procedure. The transfer residual toner includes a mixture of toner that is positively charged and toner that is negatively charged but does not have enough charges. The charges on the photosensitive drum **21** after transfer is removed by the pre-exposing unit, and by causing uniform electrical discharge from the charging roller **22**, the transfer residual toner is charged again to a negative polarity. The transfer residual toner recharged to a negative polarity by the charging portion reaches the developing portion in accordance with the rotation of the photosensitive drum **21**. Then, the surface region of the photosensitive drum **21** having passed the charging portion is exposed by the scanner unit **11** and an electrostatic latent image is drawn thereon in a state in which the transfer residual toner is still attached thereto.
(199) Here, the behavior of the transfer residual toner having reached the developing portion will be described for an exposed portion and a non-exposed portion of the photosensitive drum **21** separately. In the developing portion, the transfer residual toner attached to the non-exposed portion of the photosensitive drum **21** is transferred onto the developing roller **31** due to the potential difference between the developing voltage and the potential of the non-exposed portion of the photosensitive drum **21**, that is, the dark potential, and is collected into the developer container **32**. This is because assuming that the normal charging polarity of the toner is a negative polarity, the polarity of the developing voltage applied to the developing roller **31** is relatively positive with respect to the potential of the non-exposed portion. To be noted, the toner collected into the developer container **32** is dispersed in the toner in the developer container **32** by being agitated by the agitation member **34**, and is used for the developing process again by being born on the developing roller **31**.
(200) In contrast, the transfer residual toner attached to the exposed portion of the photosensitive drum **21** is not transferred from the photosensitive drum **21** to the developing roller **31** in the developing portion, and remains on the surface of the photosensitive drum **21**. This is because assuming that the normal charging polarity of the toner is a negative polarity, the polarity of the developing voltage applied to the developing roller **31** is further negative with respect to the potential of the exposed portion, that is, light potential. The transfer residual toner remaining on the surface of the photosensitive drum **21** is born on the photosensitive drum **21** moved to the transfer portion together with other particles of toner transferred from the developing roller **31** onto the exposed portion, and is transferred onto the recording material in the transfer portion.
(201) By employing the cleanerless configuration, a space for installing a collection container for collecting the transfer residual toner or the like becomes unnecessary, thus the size of the image forming apparatus **1** can be further reduced, and the cost of printing can be reduced by reusing the transfer residual toner.
(4) Third Modification Example
(202) Next, a third modification example in which the configuration of the process cartridge is different from any embodiments described above will be described. The present modification example has the same elements as in the first embodiment except for elements related to the process cartridge, and therefore description of the same elements will be omitted.
(203) (4-1) Third Mode of Process Cartridge
(204) FIGS. **18**A to **18**C are respectively a perspective view, a side view, and a section view of a process cartridge **20**B according to the present modification example. FIG. **18**C is a section view taken at a cutting position illustrated in FIG. **18**B.
(205) As illustrated in FIGS. **18**A to **18**C, the process cartridge **20**B of the present modification example includes the developing unit **802** and the drum unit **803**A. In contrast with the third embodiment, the toner receiving unit **801** is omitted, and the replenishment container attaching portion **701**, the first conveyance member **8013**, and the second conveyance member **8014** are

disposed in the developing unit **802**. That is, the present modification example is a configuration in which a replenishment container such as the toner pack **40** or the toner bottle unit **900** is attached to the replenishment port **8012** provided in the developer container **32** from the outside of the image forming apparatus to perform toner replenishment. The configuration of the replenishment container attaching portion **701** is the same as in the first embodiment, and FIGS. **18**A to **18**C illustrate a simplified shape thereof.
(206) The laser passing space SP serving as a gap for the laser light L emitted from the scanner unit **11** illustrated in FIG. **1**A toward the photosensitive drum **21** to pass through is defined between the developing unit **802**, the drum unit **803**A, and the toner receiving unit **801**. In addition, it is preferable that, in the drum unit **803**A, a pre-exposing unit for removing the electrostatic latent image by radiating light onto the surface of the photosensitive drum **21** is disposed downstream of the transfer portion and between the transfer portion and the charging roller **22** in the rotation direction of the photosensitive drum **21**. A cleanerless configuration is employed in the present modification example. The behavior of toner in the cleanerless configuration is the same as in the second modification example, and therefore the description thereof will be omitted.
(5) Control System of Image Forming Apparatus
(207) FIG. **19** is a block diagram illustrating a control system of the image forming apparatus **1** according to the first embodiment. The controller **90** serving as a controller of the image forming apparatus **1** includes a central processing unit: CPU **91** serving as a processing device, a random access memory: RAM **92** used as a work area of the CPU **91**, and a nonvolatile memory **93** that stores various programs. In addition, the controller **90** includes an I/O interface **94** serving as an input/output port connected to an external device, and an A/D conversion portion **95** that converts an analog signal into a digital signal. The CPU **91** reads out and executes a control program stored in the nonvolatile memory **93**, and thus controls each component of the image forming apparatus **1**. Therefore, the nonvolatile memory **93** serves as a non-transitory computer-readable recording medium storing a control program for causing an image forming apparatus to operate by a specific method.
(208) In addition, the controller **90** is connected to a T memory **57** and a P memory **58**. The T memory **57** is a nonvolatile memory included in a replenishment container such as the toner pack **40** or the toner bottle unit **900**, and the P memory **58** is a nonvolatile memory included in the process cartridge **20**. Examples of the T memory **57** serving as a storage portion provided in the replenishment container include the memory unit **45** included in the toner pack **40** described above, and the memory unit **911** included in the toner bottle unit **900** described above. In addition, the T memory **57** also stores toner information indicating that the toner stored in the replenishment container such as the toner pack **40** or the toner bottle unit **900** can be supplied to the developer container **32** for replenishment. The toner information is, for example, information describing whether or not the toner pack **40** is unused, and describing the initial amount, expiration date, and the like of the toner. In addition, the P memory **58** stores information of the remainder amount of toner accommodated in the developer container **32**, information of the total amount of toner that has been supplied from the replenishment container, information of the lifetime of the photosensitive member, information of the replacement timing of the process cartridge **20**, and the like.
(209) Further, the controller **90** is connected to a rotation locking mechanism **59** and the image forming portion **10**. Examples of the rotation locking mechanism **59** include the locking member **7014** illustrated in FIGS. **9**A to **9**C, **11**A, and **11**B provided in the replenishment container attaching portion **701** and the pressing mechanism **600** illustrated in FIG. **13** that moves the locking member **7014**. The image forming portion **10** includes a motor M**1** as a drive source that drives the photosensitive drum **21**, the developing roller **31**, the supply roller **33**, the agitation member **34**, and the like. To be noted, a single drive source does not have to be shared among these rotary members, and for example, the photosensitive drum **21**, the developing roller **31**, the supply roller

33, and the agitation member **34** may be respectively driven by different motors. In addition, the image forming portion **10** also includes a power source portion **211** for applying a voltage to each member such as the developing roller **31**, and an exposure controller **212** that controls the scanner unit **11**.
(210) A toner remainder amount detection portion **51**, a waste toner fullness detection portion **52**, an attachment detection portion **53**, an opening/closing detection portion **54**, a rotation detection portion **55**, and a push-in detection portion **56** are connected to the input side of the controller **90**.
(211) The toner remainder amount detection portion **51** detects the remainder amount of toner accommodated in the developer container **32**. Examples of the toner remainder amount detection portion **51** include the optical sensor denoted by **51***a* and **51***b* in FIG. **6**A. This optical sensor includes a light emitting portion **51***a* that emits detection light toward the inside of the developer container **32**, and a light receiving portion **51***b* that detects the detection light. In this case, the ratio of time in which the optical path of the detection light is blocked by the toner with respect to the rotation period of the agitation member **34**, that is, a Duty value, is correlated with the toner remainder amount in the developer container **32**. According to this, the toner remainder amount can be obtained from a current Duty value by preparing a correspondence relationship between the Duty value and the toner remainder amount in advance. To be noted, such an optical sensor is just an example of the toner remainder amount detection portion **51**, and alternatively a pressure sensor or an electrostatic capacitance sensor may be used. The waste toner fullness detection portion **52** detects that the amount of waste toner accumulated in the waste toner chamber **8033** of the cleaning unit **803** illustrated in FIG. **6**A has reached a predetermined upper limit. As the waste toner fullness detection portion **52**, for example, a pressure sensor disposed in the waste toner chamber **8033** can be used. In addition, the controller **90** may estimate the amount of waste toner by calculation based on the image information by assuming that a certain ratio of toner corresponding to the image information is collected as waste toner.
(212) The attachment detection portion **53** detects that a replenishment container such as the toner pack **40** is attached to the replenishment container attaching portion **701**. For example, the attachment detection portion **53** is constituted by a pressure switch that is provided in the replenishment container attaching portion **701** and outputs a detection signal when pressed by the bottom surface of the toner pack **40**. In addition, the attachment detection portion **53** may be a detection circuit that detects that the T memory **57** has been electrically connected to the controller **90** via the contact portion **70133** of the replenishment container attaching portion **701** illustrated in FIGS. **9**A to **9**C.
(213) The rotation detection portion **55** detects the rotation of the replenishment container attached to the replenishment container attaching portion **701**. Examples of the rotation detection portion **55** include the rotation detection portion **7015** constituted by the leaf springs **70151** and **70152** illustrated in FIGS. **9**A to **9**C and **16**A to **16**F. The rotation detection portion **7015** is merely an example of the rotation detection portion **55**, and alternatively, for example, a photoelectric sensor shielded by a projection portion provided on the replenishment port shutter **7013** may be used as a rotation detection sensor. In addition, as another example of the rotation detection sensor, a configuration in which the conduction between the leaf springs **70151** and **70152** of the rotation detection portion **7015** is caused by a projection portion provided on the discharge portion **42** of the toner pack **40** may be employed.
(214) The push-in detection portion **56** is an element that is additionally provided in the case of using the toner bottle unit **900** as in the first modification example, and detects completion of push-in of the piston **902** of the toner bottle unit **900**. Examples of the push-in detection portion **56** include a detection circuit that is provided in the image forming apparatus **1** and detects the change in the state of the push-in detection mechanism illustrated in FIGS. **16**A to **16**F constituted by the push-in detection rod **907**, the first contact plate **908**, and the second contact plate **909** provided in the toner bottle unit **900**. This detection circuit monitors the value of current generated when a

voltage is applied between the metal plates **9111** and **9113** respectively connected to the first contact plate **908** and the second contact plate **909**, and thus detects whether the piston **902** has been pushed in or has not been pushed in yet.

(215) In addition, the controller **90** is connected to the operation portion **300** serving as a user interface of the image forming apparatus **1**, and the panel **400** serving as a notification portion that notifies the user of information related to the toner remainder amount in the developer container **32**. Here, the information related to the toner remainder amount is not limited to information indicating the toner remainder amount itself. In addition to this, examples of the information related to the toner remainder amount include information indicating the amount of toner that has been already supplied from the toner pack **40** or the toner bottle unit **900** for replenishment. In addition, examples of the information related to the toner remainder amount include information indicating the remaining capacity of the developer container **32** that indicates the amount of toner that can be accepted by the developer container **32** for replenishment in terms of the number of toner packs **40** or toner bottle units **900**.

(216) The operation portion **300** includes a display portion **301** capable of displaying various setting screens. For example, the display portion **301** is constituted by a liquid crystal panel. In addition, the operation portion **300** includes an input portion **302** that receives an input operation from a user. For example, the input portion **302** is constituted by a physical button or a touch panel function portion of the liquid crystal panel. To be noted, the operation portion **300** may have a configuration including a sound generating portion such as a loudspeaker that notifies information related to the toner remainder amount or information related to a procedure of toner replenishment by a sound.

(217) In addition, the image forming apparatus **1** is communicably connected to information processing apparatuses such as a personal computer: PC **2**A and a mobile information processing terminal **2**B such as a smartphone as illustrated in FIG. **20**. Information transmitted to the image forming apparatus **1** from the PC **2**A and the mobile information processing terminal **2**B is input to the controller **90** through the I/O interface **94**. In addition, information transmitted from the image forming apparatus **1** to the PC **2**A or the mobile information processing terminal **2**B is input from the controller **90** to a controller of the PC **2**A or a controller of the mobile information processing terminal **2**B through the I/O interface **94**. To be noted, a configuration in which the PC **2**A and the mobile information processing terminal **2**B are provided with a sound generating portion such as a loudspeaker may be employed.

(6) Display of Panel

(218) The panel **400** serving as a second display portion displays whether or not the image forming apparatus **1** can be replenished with toner, and also displays the amount of toner that can be added for replenishment in terms of the number of toner packs **40**. For example, the process cartridge **20** is filled with 110 g of toner when the process cartridge **20** is brand-new, and 5000 ISO images can be printed with this amount of toner. For example, a brand-new toner pack **40** is filled with 50 g of toner.

(219) In the case where the amount of toner accommodated in the process cartridge **20** is 60 g to 110 g, lights of the three indicators **4001**, **4002**, and **4003** of the panel **400** are on, and the panel **400** takes a first state as illustrated in FIG. **21**A. In the case where the amount of toner accommodated in the process cartridge **20** is 10 g or larger and smaller than 60 g, the lights of the two indicators **4002** and **4003** of the panel **400** are on and the light of the uppermost indicator **4001** is off as illustrated in FIG. **21**B. That is, the panel **400** takes a second state.

(220) As described above, the panel **400** includes a plurality indicators. In the first state, lights of a first number of indicators among the plurality of indicators are on, and in the second state, lights of a second number of indicators among the plurality of indicators are on. The second number is smaller than the first number. In the present embodiment, the first number is three and the second number is two. Such a relationship of numbers of indicators that light up also applies to the panel

400 of the second state and the panel **400** of the third state.

(221) Here, in the case where the amount of toner that can be added to the developer container **32** for replenishment when the panel **400** is in the first state is a first amount, the amount of toner that can be added to the developer container **32** for replenishment when the panel **400** is in the second state is a second amount larger than the first amount. That is, the panel **400** takes the second state in the case where the amount of toner that can be added to the developer container **32** for replenishment is larger than in the first state. To be noted, examples of the first amount include 0.

(222) In the case where the amount of toner accommodated in the process cartridge **20** is larger than 0 g and smaller than 10 g, the light of the one indicator **4003** of the panel **400** is on, and the lights of the two upper indicators **4001** and **4002** are off as illustrated in FIG. **21**C. That is, the panel **400** takes a third state. In the case where the amount of toner in the process cartridge **20** is 0 g, the lights of the three indicators **4001**, **4002**, and **4003** of the panel **400** are off as illustrated in FIG. **21**D, and the panel **400** takes a fourth state. In the case where the panel **400** is in the fourth state, the image forming apparatus **1** cannot perform printing.

(223) To be noted, although a relationship between the indicators **4001** to **4003** of the panel **400** and the amount of toner is set as described above, the values of the amount of toner are not limited to these values, and can be set as appropriate. In addition, the shape of the panel **400** and the number of indicators are not limited.

(224) In the case where the process cartridge **20** is replenished with toner from the toner pack **40** when the panel **400** is in the third state or the fourth state, the panel **400** switches to the second state as illustrated in FIG. **21**B. In the case where the process cartridge **20** is replenished with toner from the toner pack **40** when the panel **400** is in the second state, the panel **400** switches to the first state as illustrated in FIG. **21**A.

(225) In the present embodiment, the amount of toner accommodated in the developer container **32** before the replenishment operation of supplying toner from the toner pack **40** to the replenishment port **8012** of the developer container **32** is performed is calculated by the controller **90** by a pixel counting method. The amount of toner accommodated in the developer container **32** before the replenishment operation will be hereinafter referred to as a pre-replenishment toner remainder amount. The pixel counting method is a method of calculating the amount of toner consumption from the number of pixels of an image formed on the recording material and obtaining the amount of toner in the developer container **32** from this amount of toner consumption. The amount of toner consumption according to the pixel counting method is obtained by multiplying the number of pixels of the photosensitive drum **21** exposed by the laser light L by the amount of toner consumption per pixel.

(226) Then, by subtracting the amount of toner consumption calculated by the pixel counting method from the amount of toner accommodated in the process cartridge **20**, the pre-replenishment toner remainder amount is calculated. The controller **90** calculates the pre-replenishment toner remainder amount each time printing is performed on a sheet, and stores the amount of toner in the P memory **58**.

(7) Control in Toner Replenishment

(227) Next, control performed by the controller **90** in toner replenishment will be described with reference to a flowchart of FIG. **22**. As illustrated in FIG. **22**, first, in step S**601**, the controller **90** recognizes that the replenishment operation from the toner pack **40** or the toner bottle unit **900** to the replenishment port **8012** has been performed.

(228) The replenishment operation performed using the toner pack **40** is determined on the basis of the elapse of a predetermined time from a time point when the rotation detection portion **7015** has been switched to the connected state as described above. In addition, the replenishment operation performed using the toner bottle unit **900** is determined on the basis of detection of completion of the push-in operation of the piston **902** by the push-in detection mechanism. An example in which the replenishment operation is performed by using the toner pack **40** will be described below.

(229) When the controller **90** detects that the replenishment operation using the toner pack **40** has been completed, the controller **90** changes the display of the panel **400** in step S**602**. For example, the controller **90** switches the panel **400** from the second state to the first state.
(230) Next, the controller **90** sums up the pre-replenishment toner remainder amount read out from the P memory **58** and the amount of toner originally accommodated in the toner pack **40**. Thus, in step S**603**, the controller **90** calculates the amount of toner accommodated in the developer container **32** after the replenishment operation. This amount will be hereinafter referred to as a post-replenishment toner remainder amount. That is, the post-replenishment toner remainder amount is the sum of the pre-replenishment toner remainder amount and the amount of toner originally accommodated in the toner pack **40**.
(231) Next, in step S**604**, the controller **90** converts the calculated post-replenishment toner remainder amount into the number of sheets on which printing can be performed before the toner is consumed and it becomes impossible to perform printing. This number will be hereinafter referred to as a printable sheet number. In other words, the controller **90** converts the calculated post-replenishment toner remainder amount into the number of sheets on which printing can be performed before the toner in the developer container **32** reaches the Out level as a result of the printing. Further, in step S**605**, the controller **90** performs a display processing of displaying the printable sheet number on the display portion **301** serving as a first display portion, and finishes the control for toner replenishment.
(232) FIG. **23** is a graph illustrating the amount of toner in the developer container **32** in the case where ISO images are successively printed. A pattern 1 indicated by a solid line in FIG. **23** represents the amount of toner in the developer container **32** in the case where the replenishment operation is performed immediately after the panel **400** is switched from the first state to the second state. A pattern 2 indicated by a broken line in FIG. **23** represents the amount of toner in the developer container **32** in the case where the replenishment operation is performed after printing is performed on 1500 sheets after the panel **400** is switched from the first state to the second state.
(233) In either case of the pattern 1 and pattern 2, the panel **400** is in the first state immediately after the replenishment operation. Therefore, even in the case where the toner amount has a value indicated by a point B of the pattern 2, since the panel **400** is in the first state, the user can misunderstand that the developer container **32** is full of toner. However, the actual amount of toner in the developer container **32** is different between the patterns 1 and 2 as indicated by points A and B in FIG. **23**, and the developer container **32** is not full of toner with the amount of toner indicated by the point B.
(234) In addition, in the pattern 2, if printing is continued after the replenishment operation, the panel **400** switches from the first state to the second state again right away. In this case, the user may misunderstand that the toner replenished from the toner pack **40** has been all consumed by the printing after the replenishment operation, which leaves the user a bad impression.
(235) Therefore, in the present embodiment, the printable sheet number is displayed on the display portion **301** after toner replenishment as described in step S**605** of FIG. **22**. For example, as illustrated in FIGS. **24**A and **24**B, in the pattern 1, a message “Now you can print on 5000 sheets.” is displayed on the display portion **301** in toner replenishment. In addition, in the pattern 2, a message “Now you can print on 3500 sheets.” is displayed on the display portion **301** after the toner replenishment.
(236) As described above, in the present embodiment, the difference in toner amount between the points A and B described with reference to FIG. **23** is supplemented with the message displayed on the display portion **301**. Particularly, in the case where the amount of toner accommodated in the developer container **32** is different although the panel **400** is in the same state, the user can grasp the correct amount of toner from the printable sheet number displayed on the display portion **301**. Therefore, the misunderstanding about the toner amount by the user derived from the display of the panel **400** can be reduced, and the usability can be improved.

(237) In addition, according to the present embodiment, a mode of an image forming apparatus can be provided.

(238) To be noted, although a case where the replenishment operation is performed when the panel **400** is in the second state is described as an example in the present embodiment, the configuration is not limited to this. For example, in the case where the replenishment operation is performed when the panel **400** is in the third state or the fourth state, the controller **90** switches the panel **400** to the second state, and performs the display processing described above.

Second Embodiment

(239) Next, a second embodiment of the present invention will be described. The second embodiment is different from the first embodiment in what is displayed on the display portion **301**. Therefore, illustration of the same elements as in the first embodiment will be omitted, or the same elements are denoted by the same reference numerals in the illustration and description thereof will be omitted.

(240) In the present embodiment, the controller **90** performs a display processing of displaying the ratio of the post-replenishment toner remainder amount with respect to the maximum amount of toner that can be accommodated in the developer container **32** on the display portion **301**. That is, the controller **90** converts the post-replenishment toner remainder amount calculated in step S**603** of FIG. **22** into a percentage with the maximum amount of toner that can be accommodated in the developer container **32** as 100%. In addition, the minimum amount of toner in the developer container **32**, that is, 0 g is converted into 0%. The amount of toner set to 0% is not limited to 0 g, and an amount of toner at which a problem occurs in an image may be set to 0%.

(241) For example, as illustrated in FIGS. **25**A and **25**B, in the pattern 1, a message “Now the toner amount is 100%.” is displayed on the display portion **301** after the toner replenishment. In addition, in the pattern 2, a message “Now the toner amount is 70%.” is displayed on the display portion **301** after the toner replenishment.

(242) As a result of this, the user can grasp the correct amount of toner in the process cartridge **20**. In addition, according to the present embodiment, a mode of an image forming apparatus can be provided.

Third Embodiment

(243) Next, a third embodiment of the present invention will be described. The third embodiment is different from the first embodiment in what is displayed on the display portion **301**. Therefore, illustration of the same elements as in the first embodiment will be omitted, or the same elements are denoted by the same reference numerals in the illustration and description thereof will be omitted.

(244) In the present embodiment, the controller **90** performs a display processing of displaying, on the display portion **301**, the number of sheets on which printing can be performed before the next replenishment. The number of sheets on which printing can be performed before the next replenishment is the number of sheets on which printing can be performed after the replenishment operation and before the state of the panel **400** switches. For example, the controller **90** displays the printable sheet number before the panel **400** switches from the first state to the second state, which corresponds to the post-replenishment toner remainder amount, on the display portion **301**. That is, the printable sheet number is the number of sheets on which printing can be performed before the toner accommodated in the developer container **32** is consumed and the panel **400** switches from the first state to the second state.

(245) For example, as illustrated in FIGS. **26**A and **26**B, in the pattern 1, a message “Now you can print on 2500 sheets before replenishment.” is displayed on the display portion **301** after the toner replenishment. In addition, in the pattern 2, a message “Now you can print on 1000 sheet before replenishment.” is displayed on the display portion **301** after the toner replenishment.

(246) As a result of this, the user can grasp the correct amount of toner in the process cartridge **20**. In addition, according to the present embodiment, a mode of an image forming apparatus can be

provided.
(247) To be noted, although the printable sheet number before the panel **400** switches from the first state to the second state is displayed on the display portion **301** in the present embodiment, the configuration is not limited to this. For example, the printable sheet number before the panel **400** switches from the first state to the third state may be displayed on the display portion **301**.
Fourth Embodiment
(248) Next, a fourth embodiment of the present invention will be described. The fourth embodiment is different from the first embodiment in the configuration of the panel **400**. Therefore, illustration of the same elements as in the first embodiment will be omitted, or the same elements are denoted by the same reference numerals in the illustration and description thereof will be omitted.
(249) An image forming apparatus **1**B according to the present embodiment includes a panel **401** serving as a second display portion as illustrated in FIG. **27**, and the panel **401** is a single panel member that is not divided. The panel **401** continuously lights up in the first state, and this indicates that the toner remainder amount in the developer container **32** is at the Full level, that is, the developer container **32** is full.
(250) The panel **401** intermittently lights up in the third state, and this indicates that the toner remainder amount in the developer container **32** is at the Low level. The light of the panel **401** is off in the fourth state, and this indicates that the toner in the developer container **32** is at the Out level. The first state, the third state, and the fourth state described above respectively correspond to the first state, the third state, and the fourth state of the panel **400** of the first embodiment.
(251) However, in the present embodiment, for example, 100 g of toner is accommodated in a brand-new process cartridge **20**. In addition, for example, 70 g of toner is accommodated in a brand-new toner pack **40**. In the case where the amount of toner accommodated in the process cartridge **20** is 30 g to 100 g, the panel **401** takes the first state. In the case where the amount of toner accommodated in the process cartridge **20** is larger than 0 g and smaller than 30 g, the panel **401** takes the third state. In the case where the amount of toner accommodated in the process cartridge **20** is 0 g, the panel **401** takes the fourth state. To be noted, although the relationship between the state of the panel **401** and the amount of toner is set as described above, the values of the amount of toner are not limited to these, and can be set appropriately.
(252) Control performed by the controller **90** in the toner replenishment is the same as in the first embodiment. That is, when the controller **90** detects that the replenishment operation using the toner pack **40** has been completed, the controller **90** changes the display of the panel **401**, and displays the printable sheet number on the display portion **301** on the basis of the calculated post-replenishment toner remainder amount.
(253) For example, as illustrated in FIGS. **28**A and **28**B, in the pattern 1, a message “Now you can print on 5000 sheets.” is displayed on the display portion **301** after the toner replenishment. In addition, in the pattern 2, a message “Now you can print on 3500 sheets.” is displayed on the display portion **301** after the toner replenishment.
(254) As a result of this, the user can grasp the correct amount of toner in the process cartridge **20**. In addition, according to the present embodiment, a mode of an image forming apparatus can be provided.
(255) To be noted, although the printable sheet number is displayed on the display portion **301** in the present embodiment similarly to the first embodiment, the configuration is not limited to this. For example, messages described in the second and third embodiments may be displayed on the display portion **301**.
(256) In addition, although the panel **401** flickers in the third state in the present embodiment, the configuration is not limited to this. For example, the panel **401** may light up at a first brightness in the first state and light up at a second brightness lower than the first brightness in the third state serving as a second state. For example, the panel **401** may light up in a first color in the first state

and light up in a second color different from the first color in the third state serving as a second state.

Other Embodiments

(257) Although the controller **90** displays a message related to the amount of toner remaining in the developer container **32** on the display portion **301** provided in the image forming apparatus in all the embodiments described above, the configuration is not limited to this. For example, as illustrated in FIG. **20**, the message may be displayed on a display portion **301** serving as a first display portion of the PC **21** or a display portion **304** serving as a first display portion of the mobile information processing terminal **2**B. In addition, the message described above may be displayed on two or more of the display portions **301**, **303**, and **304**.

(258) In addition, although the pre-replenishment toner remainder amount is calculated by using the amount of toner consumption calculated by the pixel counting method in all the embodiments described above, the configuration is not limited to this. The toner remainder amount detection portion **51** illustrated in FIGS. **6**A and **19** described above changes the output value thereof on the basis of the amount of toner accommodated in the developer container **32**. For example, the controller **90** may obtain the pre-replenishment toner remainder amount on the basis of the output value of the toner remainder amount detection portion **51**. In addition, for example, the controller **90** may obtain the pre-replenishment toner remainder amount on the basis of the output value of the toner remainder amount detection portion **51** and the amount of toner consumption calculated by the pixel counting method.

(259) In addition, although the description has been given on the premise that the amount of toner accommodated in the toner pack **40** is limited to one value in all the embodiments described above, the configuration is not limited to this. For example, a plurality of kinds of toner packs accommodating different amounts of toner may be connectable to the replenishment port **8012** of the developer container **32**. In this case, for example, the memory unit **45** illustrated in FIG. **7**A provided in the toner pack stores the amount of toner accommodated in the toner pack. The controller **90** obtains the amount of toner accommodated in the toner pack from the memory unit **45** via the contact portion **70133** of the replenishment container attaching portion **701** in contact with the memory unit **45**. In addition, the amount of toner discharged from the toner pack to the replenishment port **8012** may be detected by a sensor.

(260) In addition, although the post-replenishment toner remainder amount is calculated by summing up the pre-replenishment toner remainder amount and the amount of toner originally accommodated in the toner pack **40** in all the embodiments described above, the configuration is not limited to this. For example, in the case where the time after the toner in the toner pack **40** is discharged to the replenishment port **8012** and before the discharged toner reaches the replenishment port **8012** is short, the post-replenishment toner remainder amount may be detected by the toner remainder amount detection portion **51**.

(261) In addition, although the printable sheet number is calculated by assuming a case of printing ISO images in all the embodiments described above, the configuration is not limited to this. For example, an average amount of toner consumption per sheet may be calculated by the pixel counting method on the basis of a printing history of the user, and the printable sheet number may be obtained from this average amount of toner consumption.

(262) In addition, the message displayed on the display portion **301** in the display processing is not limited to the messages displayed on the display portion **301** in the first to fourth embodiments described above, and may be any message as long as the message indicates information related to the amount of toner accommodated in the developer container **32** after the replenishment operation.

(263) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the

above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
(264) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(265) This application claims the benefit of Japanese Patent Application No. 2019-182216, filed Oct. 2, 2019, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus to and from which a replenishment container accommodating toner is attachable and detachable and which is configured to form an image on a recording material, the image forming apparatus comprising: an image bearing member; a developer container configured to accommodate toner; a developing roller configured to develop an electrostatic image formed on the image bearing member into a toner image by using the toner accommodated in the developer container; a replenishment port configured to allow replenishment of toner from the replenishment container, which is arranged outside of the image forming apparatus, to the developer container therethrough in a state where the replenishment container is attached to the replenishment port; an opening/closing portion configured to open the replenishment port in an open position and close the replenishment port in a closed position, the opening/closing portion moving from the closed position to the open position in a case where the replenishment container attached to the replenishment port is rotated; a locking member configured to move between a restricting position in which the locking member restricts movement of the opening/closing portion from the closed position to the open position, and an allowing position in which the movement of the opening/closing portion from the closed position to the open position is allowed; a first display configured to display information related to the toner accommodated in the developer container; a second display configured to switch between a first state and a second state different from the first state, the developer container being capable of accepting more toner for replenishment in a case where the second display is in the second state than in a case where the second display is in the first state; and a controller configured to control the locking member, the first display and the second display, wherein the first display provides more information related to the toner accommodated in the developer container than the second display, and wherein the controller, (1) moves the locking member from the restricting position to the allowing position in a case where the second display is in the second state before a replenishment operation, in which toner is supplied from the replenishment container to the replenishment port, is performed, and (2) in a case where the replenishment operation is performed, performs a display processing of displaying, on the first display, the information related to the toner accommodated in the developer container after the

replenishment operation.
2. The image forming apparatus according to claim 1, further comprising a detector whose output value changes on a basis of completion of the replenishment operation, wherein the controller performs the display processing on a basis of a change of the output value of the detector.
3. The image forming apparatus according to claim 1, wherein the controller obtains an amount of toner accommodated in the developer container after the replenishment operation, from a sum of an amount of toner accommodated in the developer container before the replenishment operation and an amount of toner accommodated in the replenishment container.
4. The image forming apparatus according to claim 1, wherein the controller obtains an amount of toner accommodated in the developer container before the replenishment operation, by using an amount of toner consumption calculated from a number of pixels of an image formed on a recording material.
5. The image forming apparatus according to claim 1, further comprising a toner remainder amount detector whose output value changes on a basis of an amount of toner accommodated in the developer container, wherein the controller obtains the amount of toner accommodated in the developer container before the replenishment operation, on a basis of the output value of the toner remainder amount detector.
6. The image forming apparatus according to claim 1, further comprising a toner remainder amount detector whose output value changes on a basis of an amount of toner accommodated in the developer container, wherein the controller obtains the amount of toner accommodated in the developer container before the replenishment operation, on a basis of the output value of the toner remainder amount detector and an amount of toner consumption calculated from a number of pixels of an image formed on a recording material.
7. The image forming apparatus according to claim 1, wherein in a case where the replenishment operation is performed when the second display is in the second state, the controller switches the second display from the second state to the first state.
8. The image forming apparatus according to claim 1, wherein the second display comprises a plurality of indicators, a first number of indicators among the plurality of indicators light up in a state where the second display is in the first state, and a second number of indicators among the plurality of indicators light up in a state where the second display is in the second state, the second number being smaller than the first number.
9. The image forming apparatus according to claim 1, wherein the second display is a panel member that continuously lights up in the first state and intermittently lights up in the second state.
10. The image forming apparatus according to claim 1, wherein the second display is a panel member that lights up at a first brightness in the first state and lights up at a second brightness lower than the first brightness in the second state.
11. An image forming apparatus to and from which a replenishment container accommodating toner is attachable and detachable and which is communicable with an information processing apparatus comprising a first display and is configured to form a toner image on a recording material, the image forming apparatus comprising: an image bearing member; a developer container configured to accommodate toner; a developing roller configured to develop an electrostatic image formed on the image bearing member into a toner image by using the toner accommodated in the developer container; a replenishment port configured to allow replenishment of toner from the replenishment container, which is arranged outside of the image forming apparatus, to the developer container therethrough in a state where the replenishment container is attached to the replenishment port; an opening/closing portion configured to open the replenishment port in an open position and close the replenishment port in a closed position, the opening/closing portion moving from the closed position to the open position in a case where the replenishment container attached to the replenishment port is rotated; a locking member configured to move between a restricting position in which the locking member restricts movement of the

opening/closing portion from the closed position to the open position, and an allowing position in which the movement of the opening/closing portion from the closed position to the open position is allowed; a second display configured to switch between a first state and a second state different from the first state, the developer container being capable of accepting more toner for replenishment in a case where the second display is in the second state than in a case where the second display is in the first state; and a controller configured to control the locking member, the first display and the second display, wherein the first display provides more information related to the toner accommodated in the developer container than the second display, and wherein the controller (1) moves the locking member from the restricting position to the allowing position in a case where the second display is in the second state before a replenishment operation, in which toner is supplied from the replenishment container to the replenishment port, is performed, and (2) in a case where the replenishment operation is performed, performs a display processing of displaying, on the first display, the information related to the toner accommodated in the developer container after the replenishment operation.

12. The image forming apparatus according to claim 11, further comprising a detector whose output value changes on a basis of completion of the replenishment operation, wherein the controller performs the display processing on a basis of a change of the output value of the detector.

13. The image forming apparatus according to claim 11, wherein the second display comprises a plurality of indicators, a first number of indicators among the plurality of indicators light up in a state where the second display is in the first state, and a second number of indicators among the plurality of indicators light up in a state where the second display is in the second state, the second number being smaller than the first number.

14. The image forming apparatus according to claim 11, wherein the second display is a panel member that continuously lights up in the first state and intermittently lights up in the second state.

15. The image forming apparatus according to claim 11, wherein the second display is a panel member that lights up at a first brightness in the first state and lights up at a second brightness lower than the first brightness in the second state.

16. The image forming apparatus according to claim 11, wherein the controller obtains an amount of toner accommodated in the developer container after the replenishment operation, from a sum of an amount of toner accommodated in the developer container before the replenishment operation and an amount of toner accommodated in the replenishment container.

17. The image forming apparatus according to claim 11, wherein the controller obtains an amount of toner accommodated in the developer container before the replenishment operation, by using an amount of toner consumption calculated from a number of pixels of an image formed on a recording material.

18. The image forming apparatus according to claim 11, further comprising a toner remainder amount detector whose output value changes on a basis of an amount of toner accommodated in the developer container, wherein the controller obtains the amount of toner accommodated in the developer container before the replenishment operation, on a basis of the output value of the toner remainder amount detector.

19. The image forming apparatus according to claim 11, further comprising a toner remainder amount detector whose output value changes on a basis of an amount of toner accommodated in the developer container, wherein the controller obtains the amount of toner accommodated in the developer container before the replenishment operation, on a basis of the output value of the toner remainder amount detector and an amount of toner consumption calculated from a number of pixels of an image formed on a recording material.

20. The image forming apparatus according to claim 11, wherein in a case where the replenishment operation is performed when the second display is in the second state, the controller switches the second display from the second state to the first state.

21. The image forming apparatus according to claim 11, wherein the controller, in a case where the

replenishment operation is performed, performs a display processing of displaying, on the first display, a printable sheet number corresponding to an amount of toner accommodated in the developer container after the replenishment operation.

22. The image forming apparatus according to claim 21, wherein the printable sheet number is a number of sheets on which it is possible to perform printing before the toner accommodated in the developer container is consumed and printing becomes impossible.

23. The image forming apparatus according to claim 21, wherein the printable sheet number is a number of sheets on which it is possible to perform printing before the toner accommodated in the developer container is consumed and the first display switches from the first state to the second state.

24. The image forming apparatus according to claim 21, wherein in a case where the replenishment operation is performed when the second display is in the second state, the controller switches the second display from the second state to the first state.
